

Bacterial cGAS-like enzymes synthesize diverse nucleotide signals

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Cyclic dinucleotides (CDNs) have central roles in bacterial homeostasis and virulence by acting as nucleotide second messengers. Bacterial CDNs also elicit immune responses during infection when they are detected by pattern-recognition receptors in animal cells. Here we perform a systematic biochemical screen for bacterial signalling nucleotides and discover a large family of cGAS/DncV-like nucleotidyltransferases (CD-NTases) that use both purine and pyrimidine nucleotides to synthesize a diverse range of CDNs. A series of crystal structures establish CD-NTases as a structurally conserved family and reveal key contacts in the enzyme active-site lid that direct purine or pyrimidine selection. CD-NTase products are not restricted to CDNs and also include an unexpected class of cyclic trinucleotide compounds. Biochemical and cellular analyses of CD-NTase signalling nucleotides demonstrate that these cyclic di- and trinucleotides activate distinct host receptors and thus may modulate the interaction of both pathogens and commensal microbiota with their animal and plant hosts.

Second messenger molecules allow cells to amplify signals and rapidly control downstream responses. This concept is exemplified in human cells, where mislocalized double-stranded (ds)DNA stimulates the cytosolic enzyme cyclic GMP-AMP synthase (cGAS) to synthesize the CDN 2'-5'/3'-5'-cyclic GMP-AMP (2',3'-cGAMP)^{1,2}. 2',3'-cGAMP diffuses throughout the cell, activates the receptor stimulator of interferon genes (STING) and induces type-I interferon and NF- κ B responses to elicit protective anti-viral immunity¹. Recently, synthetic CDN analogues have emerged as promising lead compounds for immune modulation and cancer immunotherapy^{2,3}.

CDNs were first identified in bacteria⁴ and this discovery established the foundation for later recognition of the importance of CDN signalling in mammalian cells⁵. Nearly all bacterial phyla encode CDN signalling pathways; however, all known natural CDN signals are constructed only from purine nucleotides⁶. CDNs control diverse responses in bacterial cells. For example, cyclic di-GMP coordinates the transition between planktonic and sessile growth, cyclic di-AMP controls osmoregulation, cell-wall homeostasis and DNA-damage responses, and 3'-5'/3'-5'-cGAMP (3',3'-cGAMP) modulates chemotaxis, virulence and exoelectrogenesis⁷. The human receptor STING also senses these bacterial CDNs as pathogen (or microbe)-associated molecular patterns (PAMPs), revealing a direct, functional connection between bacterial and human nucleotide signalling⁸. However, our understanding of the true scope of immune responses to signalling nucleotide products from bacteria is limited to cyclic dipurine molecules. Here we describe a systematic approach to understanding the diversity of products synthesized by a family of microbial synthases related to the *Vibrio cholerae* enzyme dinucleotide cyclase in *Vibrio* (DncV) and its metazoan homologue cGAS⁹⁻¹¹.

Discovery of a pyrimidine-containing CDN

The enzyme DncV synthesizes 3',3'-cGAMP and controls a signalling network on the *Vibrio* seventh pandemic island-I (VSP-I), a horizontally acquired genetic element present in all current *V. cholerae*

pandemic isolates¹¹⁻¹³. While investigating homologues of *dncV* outside the Vibrionales, we identified an unexpected partial operon in *Escherichia coli* where *dncV* is replaced with a gene of unknown function (WP_001593458, here named *cdnE*). The operon architecture suggests that *cdnE* may be an alternative CDN synthase (Fig. 1a). We tested this hypothesis by incubating purified CdnE protein with α^{32} P-radiolabelled ATP, CTP, GTP and UTP and visualized the reaction products using thin-layer chromatography (TLC). CdnE synthesized a product distinct from currently known CDNs (Fig. 1b and Extended Data Fig. 1a, b). Notably, biochemical deconvolution using pairwise assessment of necessary NTPs revealed that ATP and UTP were necessary and sufficient for product formation (Fig. 1c). We analysed the purified product using nuclease digestion, mass spectrometry and NMR (Fig. 1d and Extended Data Fig. 1d-l), and confirmed that the product of CdnE is cyclic UMP-AMP (cUMP-AMP), a hybrid purine-pyrimidine CDN.

DncV is a structural homologue of cGAS, and each enzyme uses a single active site to sequentially form two separate phosphodiester bonds and release a CDN product¹⁰. In spite of no overall sequence homology, careful inspection of the CdnE sequence revealed potential cGAS/DncV-like active-site residues (GSYX₁₀DVD), which were essential for catalysis (Extended Data Fig. 1c). Reactions with non-hydrolysable nucleotides confirmed that CdnE catalyses the synthesis of cUMP-AMP using a sequential path through a pppU[3'-5']pA intermediate (Extended Data Fig. 2) and revealed that CdnE is probably a divergent enzyme that is ancestrally related to cGAS and DncV. We therefore renamed the gene cGAS/DncV-like nucleotidyltransferase in *E. coli* (*cdnE*).

DncV controls the activity of cGAMP-activated phospholipase in *Vibrio* (CapV), a patatin-like lipase that is a direct 3',3'-cGAMP receptor encoded in the *dncV* operon¹⁴. *cdnE* is also preceded by a gene that encodes a patatin-like phospholipase (which we named cUMP-AMP activated phospholipase in *E. coli* (*capE*); Fig. 1a). We confirmed that CapE is activated by cUMP-AMP, and determined that CapV and CapE

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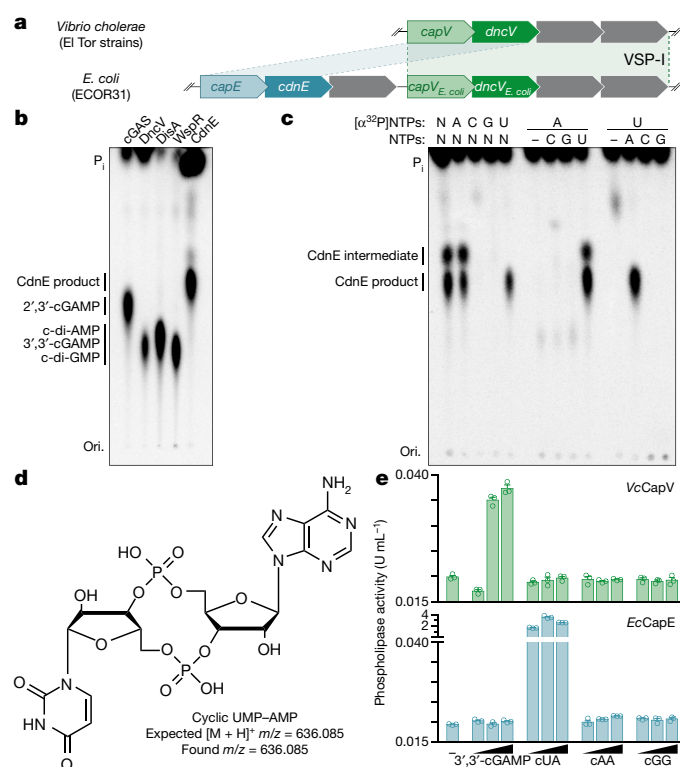


Fig. 1 | Bacteria synthesize cyclic UMP-AMP. **a**, An *E. coli* genomic island homologous to *Vibrio* seventh pandemic island-I (VSP-I) encodes a 3',3'-cGAMP synthase (*dncV*) and phospholipase receptor (*capV*). The ECOR31 strain's genomic island encodes a second *capV*-like gene (BHF03_01995 (*capE*), which encodes WP_001593459) adjacent to a gene of unknown function (BHF03_01990 (*cdnE*) which encodes WP_001593458). **b**, PEI-cellulose TLC of enzyme reactions. Purified CdnE was incubated with α^{32} P-radiolabelled ATP, CTP, GTP and UTP. Reactions were terminated by treating with alkaline phosphatase to remove exposed triphosphates. Standards are 2',3'-cGAMP (cGAS), 3',3'-cGAMP (DncV), c-di-AMP (DisA) and c-di-GMP (WspR). Data are representative of three independent experiments. P_i, inorganic phosphate; Ori., origin. **c**, Biochemical deconvolution of the CdnE reactions as in **b** after incubation with α^{32} P-labelled and unlabelled NTPs. Data are representative of three independent experiments. **d**, The CdnE product was confirmed by mass spectrometry and NMR (see Extended Data Fig. 1e–l). **e**, Activation of CapV and CapE by CDNs, tested with no nucleotide added (–) or at 0.1-, 1- and 10-fold molar ratios of nucleotide to phospholipase. Enzyme activity is reported in phospholipase A₁ units (U) ml⁻¹. Data are mean $\pm$ s.e.m. for $n = 3$ technical replicates and are representative of three independent experiments.

are specifically activated by only the nucleotide synthesized from their adjacently encoded nucleotidyltransferase (Fig. 1e). Notably, the identification of CapE as a direct cUMP-AMP receptor in *E. coli* confirms that CdnE produces cUMP-AMP to control downstream signalling. The specificity of CapE insulates this circuit from 3',3'-cGAMP and other parallel CDN signals, potentially explaining an evolutionary advantage of cUMP-AMP and increased CDN diversity.

Mechanism of pyrimidine discrimination

We determined a series of X-ray crystal structures of a CdnE homologue from the thermophilic bacterium *Rhodothermus marinus* (*RmCdnE*; Fig. 2, Extended Data Fig. 3a and Supplementary Table 1). CdnE adopts a Pol- β -like nucleotidyltransferase fold that is highly similar to cGAS and the core of DncV, confirming a shared structural and evolutionary relationship (Fig. 2d). CdnE is more distantly related to other nucleotidyltransferases, including non-templated CCA-adding enzymes, poly(A) polymerases and templated polymerases, such as DNA polymerase β and μ . The human innate immune enzymes cGAS and oligoadenylate synthase 1 (OAS1) are activated through a

conformational change induced by binding to a double-stranded nucleic acid¹⁵. CdnE, like DncV¹⁰, is structurally more similar to the 'activated' conformation of these two enzymes, consistent with biochemical analyses that demonstrate that CdnE is constitutively active and does not require a cognate stimulus in vitro.

The structure of *RmCdnE* in complex with non-hydrolysable ATP and UTP reveals an asparagine side chain (Asn166) that forms hydrogen bonds with the uracil base and positions the ATP α P for attack by the 3'-hydroxyl of UTP (Fig. 2a). Asn166 is located in the same position as a serine residue in the acceptor nucleotide pocket of both DncV and cGAS (Extended Data Fig. 3b, d), and we hypothesized that this asparagine substitution might be sufficient to dictate CdnE product specificity. Whereas wild-type CdnE robustly synthesized cUMP-AMP, CdnE^{N166S} incorporated almost no UTP and instead predominantly synthesized c-di-AMP (Fig. 2c and Extended Data Fig. 3c). We surveyed CdnE homologues and determined that Asn166 is nearly universally conserved (Fig. 2b and Extended Data Fig. 4a). An exception is CdnE from the emerging nosocomial pathogen *Elizabethkingia meningoseptica*¹⁶ (*EmCdnE*, Fig. 2b), which encodes a serine at the analogous position to Asn166. In contrast to the other CdnE homologue, *EmCdnE* robustly synthesized cyclic dipurine products (Fig. 2c and Extended Data Fig. 4b–h). Crystal structures of *EmCdnE* bound to its nucleotide substrates demonstrated natural N to S reprogramming in the active-site lid, and re-introduction of the ancestral asparagine at this position reverted *EmCdnE* back to preferential production of pyrimidine-containing products (Fig. 2c, Extended Data Fig. 4f–i and Supplementary Table 1). These data reveal a remarkably low barrier for altering specificity of CdnE and demonstrate that organisms, such as *E. meningoseptica*, harbour mutations at Asn166 that reprogram purine and pyrimidine product specificity.

High-resolution structures of cGAS, OAS1, DncV and two CdnE homologues allowed for the rational definition of shared structural and functional homology. All of these enzymes share three features: (1) a common DNA polymerase β -like nucleotidyltransferase superfamily protein fold in spite of large sequence divergence; (2) template-independent synthesis of a diffusible molecule through caging of the active site, using a protein scaffold not conserved with more distantly related templated polymerases; and (3) an active site architecture that allows diversification of products and phosphodiester linkage through amino acid substitutions within the active-site lid. We designated this family of enzymes CD-NTases (cGAS/DncV-like nucleotidyltransferases), a structurally and evolutionarily distinct subset of the DNA polymerase β -like nucleotidyltransferase superfamily (Fig. 2d). CD-NTases use distinct enzymatic chemistry and are not structurally related to dimeric GGDEF family c-di-GMP synthases or DAC/DisA family c-di-AMP synthases^{17,18}, and therefore represent a third family of CDN synthase.

CD-NTases and cross-kingdom signalling

Many bacteria that encode CD-NTases thrive in close proximity to eukaryotic hosts, including humans, plants and fungi (Fig. 2b). CdnE homologues are found in the intracellular pathogen *Shigella sonnei* and commensal genera such as *Bacteroides* (Fig. 2b). Mammals have evolved a sophisticated surveillance system for detecting and initiating immune responses to bacterial products, including CDNs that are secreted or released during bacteriolysis¹⁹. STING detects bacterial c-di-AMP, c-di-GMP and 3',3'-cGAMP in addition to endogenously produced 2',3'-cGAMP¹. We determined whether cUMP-AMP was also recognized by STING or other receptors of the innate immune system. Mouse STING bound to all four cyclic dipurine molecules with high affinity and activated type-I interferon signalling in cells. However, STING was unable to recognize cUMP-AMP in vitro at concentrations known to be sufficient for cyclic dipurine agonists and cUMP-AMP failed to activate STING-dependent type-I interferon signalling in cells (Fig. 3a and Extended Data Fig. 5a–d). These data are consistent with previous experiments using chemically synthesized nucleotides²⁰ and were not due to differences in CD-NTase expression levels (Extended Data Fig. 5c, d). By contrast, the recently described mammalian CDN

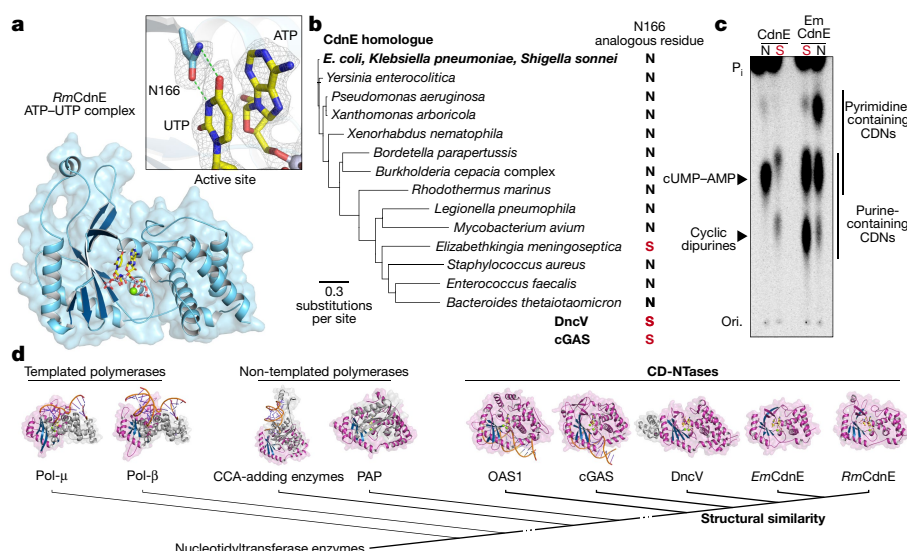


Fig. 2 | Conserved active-site residues dictate CD-NTase specificity.

a, Crystal structure of *RmCdnE* in complex with non-hydrolysable ATP and UTP analogues and magnified inset of key Asn166–uridine contacts that control pyrimidine specificity. Green dotted lines indicate hydrogen bonds and the $2F_o - F_c$ electron density map is contoured at 1σ . **b**, Phylogram of CdnE sequence homologues and their Asn166 analogous residue determined by sequence alignment (Extended Data Fig. 4a). Residues indicated in red (S) highlight cGAS/DncV-like serine residues. **c**, CdnE homologue and mutant reactions analysed by TLC as in Fig. 1b. N and red S indicate asparagine or cGAS/DncV-like serine residues at the Asn166 analogous position in the tested allele. Side chains are numbered

according to the *RmCdnE* sequence. Data are representative of three independent experiments. For detailed deconvolution and purine versus pyrimidine migration pattern analysis see Extended Data Figs. 3, 4. **d**, Structure-based comparisons define that *RmCdnE* and *EmCdnE* are CD-NTases with a similar architecture to DncV (Protein Data Bank (PDB) accession 4TY0), cGAS (PDB 6CTA) and OAS1 (PDB 4RWO). CD-NTases are more distantly related to Pol-β-like NTases: Pol-μ (PDB 4YD1), Pol-β (PDB 4KLQ), CCA-adding enzyme (PDB 4X4T) and poly(A) polymerase γ (PAP, PDB 4LT6). The NTase core domain for each enzyme is clustered according to z-score and coloured in magenta and blue.

sensor reductase controlling NF-κB (RECON)²¹ was capable of recognizing cUMP–AMP, and cUMP–AMP inhibited RECON function, albeit with a reduced potency compared to the previously reported inhibition by c-di-AMP and 3',3'-cGAMP (Fig. 3b and Extended Data Fig. 5e–g). RECON bound to cUMP–AMP with a similar low micromolar dissociation constant to that of STING for c-di-GMP⁸ (Extended Data Fig. 5g); thus, we identified a host receptor for a naturally occurring purine–pyrimidine hybrid CDN. Whereas the specificity of STING for CDNs is dependent on the presence of two purine bases, RECON requires only the minimal presence of an adenine base in a 3'–5'-linked cyclic oligonucleotide. These results highlight how discovery of natural bacterial signalling molecules can refine our understanding of host receptor specificity and demonstrate that the host response may be tuned via multiple receptors that compete for CDNs using distinct rules of engagement.

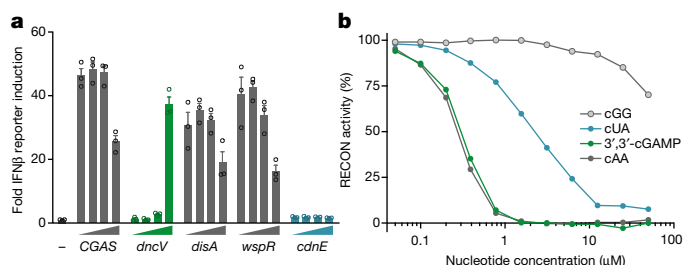


Fig. 3 | Immune detection of a pyrimidine-containing CDN. **a**, In-cell STING reporter assay. Induction of an interferon-β (IFNβ) reporter in HEK293T cells transfected with a concentration gradient of plasmid-overexpressing enzymes as indicated. cGAS synthesizes 2',3'-cGAMP, DncV synthesizes 3',3'-cGAMP, DisA synthesizes c-di-AMP, WspR synthesizes c-di-GMP and CdnE synthesizes cUMP–AMP. Data are fold induction over vector only (–). Data are mean ± s.e.m. for $n = 3$ technical replicates and are representative of three independent experiments. **b**, Nucleotide inhibition of RECON enzymatic activity, as measured by the oxidation of the NADPH co-substrate. The x axis is a log scale and data are representative of three independent experiments.

CD-NTases synthesize diverse nucleotide products

DncV and CdnE are likely to have evolved from a common ancestor, but exhibit marked divergence in primary amino acid sequence. We hypothesized that these enzymes comprise only a small fraction of existing bacterial CD-NTase diversity, and that kingdom-wide analysis of the protein family would allow systematic identification of bacterial signalling nucleotides as well as agonists/antagonists of the innate immune system. We therefore coupled bioinformatics analysis with a large-scale, forward biochemical screen to directly uncover additional nucleotide products. Previously, a hidden Markov model was derived from cGAS and DncV and conserved operon structures to identify potentially related bacterial proteins²². Building upon this previous analysis, we identified more than 5,600 unique bacterial enzymes predicted to share common CD-NTase structural features (Fig. 4a, Extended Data Fig. 6a and Supplementary Table 2). CD-NTases were identified in over 10% of bacterial genomes available in the NCBI database, within taxa that span nearly every bacterial phylum (Extended Data Fig. 6b). Bacteria harbouring CD-NTase genes include human commensal organisms (for example, Clostridiales and Fusobacteria), human pathogens (for example, *Listeria*, *Shigella* and *Salmonella* species), extremophiles and agriculturally important bacteria (for example, rhizobia and plant pathogens, such as *Xanthomonas*). Although CD-NTases are found in many different organisms, they are typically not encoded in the core genome and are encoded by specific strains from each species. Sequence alignments revealed that CD-NTases cluster into roughly eight clades that we designated A–H starting with A for the DncV-harboring clade, E for the CdnE-containing clade and continued to the letter H. We further divided highly related sequences into clusters, which often grouped bacterial species that occupy a similar niche, such as the plant rhizobia in cluster G10 (Fig. 4a and Supplementary Table 2).

We purified 66 CD-NTase proteins and tested each for nucleotide product synthesis (Fig. 4a and Extended Data Fig. 6c–g). These proteins were selected as type enzymes from each cluster based on the relevance of the organism from which they were isolated (pathogens, commensals, and bacteria predicted to interact with eukaryotes) and

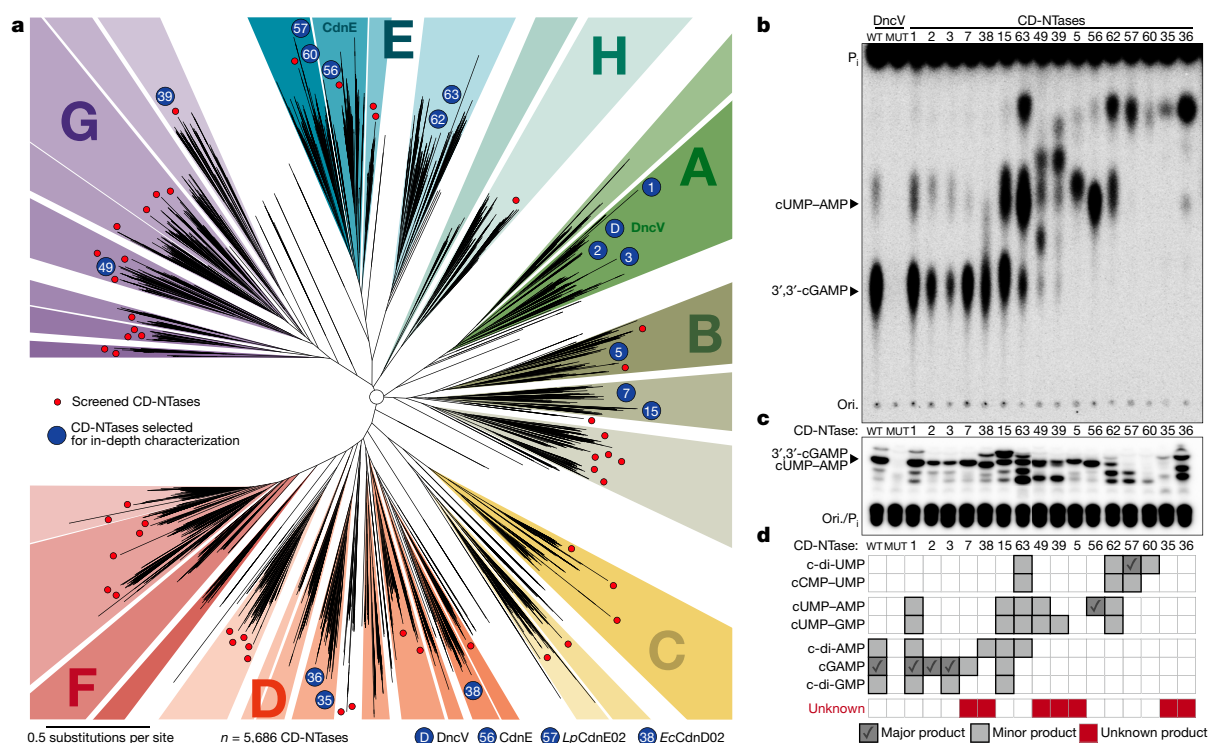


Fig. 4 | CD-NTases synthesize diverse nucleotide products and form a family of enzymes abundant in many bacterial phyla. **a**, Bioinformatic identification and clustering of ~5,600 predicted CD-NTases found in nearly every bacterial phylum, shown as an unrooted tree. Sequence-related enzymes that are approximately 10% identical are grouped by lettered clade and are similarly coloured, enzymes that are about 25% identical are grouped by cluster in a similar colour shade. Circles represent CD-NTases 001–066 that were selected as type CD-NTases for a biochemical screen (see Extended Data Fig. 6 for additional details). Blue circles denote CD-NTases selected for in-depth characterization and are labelled with CD-NTase numbers or letters from the biochemical screen (see b, CdnE is '56' and DncV is 'D'). For additional information,

see Supplementary Discussion and Supplementary Table 2. Source Data are provided in the online version of the paper. **b**, **c**, PEI-cellulose and silica TLC analysis of 16 CD-NTases selected for in-depth characterization. Activity was analysed with α - 32 P-radiolabelled NTPs as in Fig. 1b. Wild-type (WT) and catalytically inactive mutant (MUT) DncV reactions are included as controls. Screened CD-NTases were numbered CD-NTases 001–066. CD-NTase056 is CdnE, CD-NTase057 is LpCdnE02 and CD-NTase038 is EcCdnD02. These appear as '56', '57' and '38' in a, respectively. Data are representative of three independent experiments. **d**, Identification of CD-NTase products by combining TLC and mass spectrometry data. CD-NTases that synthesize a major product that could not be matched to a predicted CDN are denoted as 'unknown'.

the frequency at which each sequence has been isolated from multiple organisms. Recombinant proteins were screened using a broad range of reaction conditions to identify robust activity. Despite encoding an intact active site, no activity was observed from any representative of some CD-NTase clusters. These enzymes may function similarly to human cGAS and OAS1 for which a cognate ligand (for example, dsDNA and dsRNA) is required to stimulate enzyme activity, or it is possible that these clusters may utilize building blocks other than ribonucleotide triphosphates for product synthesis.

The 16 most active CD-NTases were selected for in-depth analysis (Fig. 4b, c). Our previous results established that cyclic dipurine and cyclic purine-pyrimidine hybrid molecules migrate at the bottom and middle of polyethylenimine (PEI)-cellulose TLC plates, respectively. In the collection of active CD-NTase representatives, several enzymes produced products that migrated at the top of the plate, even more rapidly than cUMP-AMP. Further biochemical analysis of CD-NTase057 from *Legionella pneumophila* (strain 12_4117) (WP_042646516, here named LpCdnE02) demonstrated that this class of PEI-cellulose TLC species corresponded to cyclic dipyrimidines, and LpCdnE02 synthesized predominantly c-di-UMP (Fig. 4b–d and Extended Data Fig. 7). LpCdnE02 also harbours an asparagine residue analogous to Asn166 of RmCdnE, a feature found in nearly all CD-NTases in clade E but not found in other clades. Mass spectrometry of each CD-NTase reaction coupled with the NTP substrate-dependency profile and TLC data allowed us to identify the products produced by different CD-NTases and estimate their abundance (Fig. 4d). The 16 active, representative enzymes produced 7 purine, pyrimidine, and purine-pyrimidine hybrid CDN combinations,

demonstrating that CD-NTase enzymes synthesize a diverse array of bacterial nucleotide signals (Fig. 4d).

CD-NTases are encoded in conserved operons

A unifying characteristic of almost all CD-NTase-encoding genes is their location within similar operons in predicted mobile genetic elements (Extended Data Fig. 8a). Often genes that encode identical CD-NTase proteins are found in specific strains of unrelated bacterial species, reflecting that these genes are members of the 'mobilome' (for detail on identifying CD-NTases within an organism of interest, see Supplementary Discussion). The horizontal acquisition of CD-NTases suggests that they are likely to provide a selective advantage by altering bacterial physiology through adjacent encoded receptors, similar to *capV-dncV* and *capE-cdnE*, rather than integrating into species-specific nucleotide signalling networks. It was previously demonstrated that genes adjacent to CD-NTases are effector-like and are generally involved in biological conflict, including phospholipases, nucleases and pore-forming agents^{14,22}. Coexpression of *dncV* and *capV* is toxic to *E. coli*¹⁴ and we tested whether coexpression of each CD-NTase with its adjacently encoded, putative receptor was also toxic to *E. coli*. Expression of *dncV-capV* was uniquely able to inhibit colony formation, whereas other CD-NTase-predicted receptor pairs, including the *cdnE-capE* pair, did not impair bacterial growth (Extended Data Fig. 8b, c). It is unclear whether CD-NTases are constitutively active in vivo or exhibit regulated enzymatic activity similar to the metazoan second messenger synthase cGAS. These findings demonstrate that phenotypes observed with *Vibrio dncV-capV* may not be indicative of general CD-NTase function, and that CD-NTase-containing islands

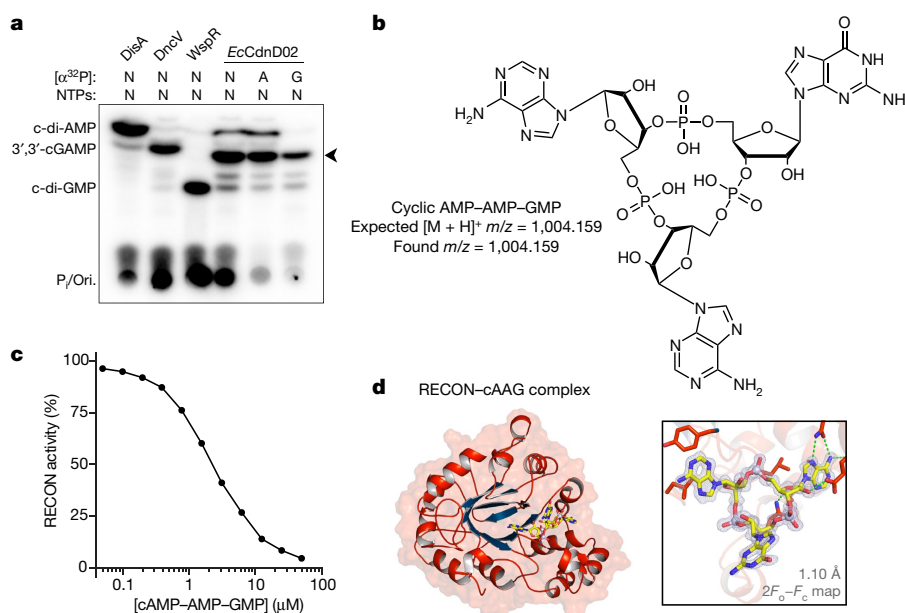


Fig. 5 | Bacterial synthesis and host recognition of a cyclic trinucleotide.

a, Silica TLC analysis of the *EcCdnD02* (CD-NTase038) product and control reactions as in Fig. 1b. The major product of *EcCdnD02* is indicated with an arrowhead and incorporates approximately $2 \times$ greater α^{32} P signal from ATP than α^{32} P signal from GTP. Data are representative of three independent experiments. **b**, The major product of *EcCdnD02*, cAAG, confirmed by mass spectrometry and NMR. See Extended Data Fig. 9 for additional characterizations. **c**, cAAG inhibition of RECON

enzymatic activity, as measured by the oxidation of the NADPH co-substrate. The x axis is a log scale and data are representative of three independent experiments. **d**, Co-crystal structure of the host receptor RECON in complex with cAAG. The inset highlights the cAAG $2F_o - F_c$ electron density contoured at 1.3σ . Green dotted lines indicate hydrogen bonds, some RECON-cAAG contacts are omitted for clarity, see Extended Data Fig. 10.

may perform varied functions such as mediating bacteriophage resistance, modulating bacteria–host interactions, functioning as addiction modules or regulating bacteriolysis for dissemination of mobile genetic elements.

CD-NTase products include cyclic trinucleotides

Notably, we were unable to identify expected CDNs by mass spectrometry in some reactions despite visualizing robust product formation by PEI–cellulose TLC. Using orthogonal silica TLC conditions, these unknown products exhibited distinct migration patterns that suggested existence of unique non-CDN species (Fig. 4c, d). We focused on an orphan product of CD-NTase038 from the *Enterobacter cloacae* complex (strain UCI 50) (WP_032676400, here named *EcCdnD02*) for identification. The *EcCdnD02* product initially appeared to be a cyclic dipurine by PEI–cellulose TLC, but the major *EcCdnD02* product displayed a unique migration pattern when analysed by silica TLC (Fig. 5a and Extended Data Fig. 9a). ATP and GTP were necessary and sufficient for product formation; however, roughly two thirds of the total α^{32} P was incorporated from ATP and the remaining third from GTP. Consistent with this pattern, evaluation of mass spectrometry data and subsequent biochemical and NMR validation revealed that cyclic AMP-AMP-GMP (cAAG), a cyclic trinucleotide, is the major product of *EcCdnD02* (Fig. 5b and Extended Data Fig. 9b–j).

Similar to cUMP-AMP, the bacterial cyclic trinucleotide cAAG escaped STING recognition but was detected by RECON, confirming our new definition of STING and RECON ligand specificity (Fig. 5c and Extended Data Fig. 10a, f). We next determined a co-crystal structure of mouse RECON in complex with the *EcCdnD02* cyclic trinucleotide product (Fig. 5d, Extended Data Fig. 10 and Supplementary Table 1). The structure further confirms that the bacterial cyclic trinucleotide is cAAG and exclusively contains 3'–5'-phosphodiester linkages. The two adenine bases are coordinated in the same adenine and nicotinamide pockets observed in the previous structure of RECON bound to bacterial c-di-AMP²¹; however, RECON Glu28 makes additional contacts with the third guanine base of the cAAG species as part of an extended base platform that is not required for

CDN recognition. Glu28 is highly conserved, potentially indicating that RECON may have evolved to enable recognition of additional bacterial or host cyclic trinucleotide species. This unexpected class of nucleotide products reveals that the active site of CD-NTase enzymes can be adapted to synthesize larger cyclic oligonucleotide products, and that host immune receptors are capable of recognizing bacterial cyclic trinucleotide species. Recently, cyclic oligoadenylates synthesized by Cas10 were demonstrated to be a key signalling molecule in type-III CRISPR immunity^{23,24}. Although CD-NTases have no homology with Cas10, these parallel findings indicate that larger cyclic oligonucleotide products may be more common in bacterial signalling and host recognition than previously expected.

CD-NTases in health and disease

Our data demonstrate that bacterial CD-NTases are widespread and synthesize diverse CDNs that include pyrimidine nucleotides and additional cyclic trinucleotide compounds. CD-NTases join the GGDEF and DAC/DisA domains, responsible for c-di-GMP and c-di-AMP synthesis^{17,18}, as a third major family of enzymes that control downstream signalling using CDN signals. Distinguishing features of CD-NTases are their location on mobile genetic elements, extreme sequence diversity and reaction mechanism that is reliant on a monomeric enzyme active site. Recent evidence demonstrates that divergent GGDEF family enzymes produce 3',3'-cGAMP in addition to c-di-GMP^{25,26}, suggesting that the selective pressures driving CD-NTase diversity may also be in effect for GGDEF and DAC/DisA-like synthases.

Understanding the functional role of CD-NTase genes in the biology of bacteria and host–microorganism interactions is a major challenge for future studies. Mammalian receptors recognize diverse CD-NTase products and CD-NTase genes may provide a selective advantage for some bacterium–eukaryote interactions. Our data show that a single mutation in a CD-NTase enables incorporation of pyrimidines and indicate that bacteria may evade or enhance STING signalling by modulating enzyme specificity. The possibility that diverse CDNs and related nucleotide signals produced by prokaryotic CD-NTases act as agonists and inhibitors of innate immunity and other host metabolic

pathways provides an important new reservoir of compounds with biotechnological and therapeutic applications.

Online content

Any methods, additional references, Nature Research reporting summaries, source data, statements of data availability and associated accession codes are available at <https://doi.org/10.1038/s41586-019-0953-5>.

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- Wu, J. & Chen, Z. J. Innate immune sensing and signaling of cytosolic nucleic acids. *Annu. Rev. Immunol.* **32**, 461–488 (2014).
- Corrales, L. et al. Direct activation of STING in the tumor microenvironment leads to potent and systemic tumor regression and immunity. *Cell Rep.* **11**, 1018–1030 (2015).
- Fu, J. et al. STING agonist formulated cancer vaccines can cure established tumors resistant to PD-1 blockade. *Sci. Transl. Med.* **7**, 283ra52 (2015).
- Ross, P. et al. Regulation of cellulose synthesis in *Acetobacter xylinum* by cyclic diguanylic acid. *Nature* **325**, 279–281 (1987).
- Danilchanka, O. & Mekalanos, J. J. Cyclic dinucleotides and the innate immune response. *Cell* **154**, 962–970 (2013).
- Nelson, J. W. & Breaker, R. R. The lost language of the RNA world. *Sci. Signal.* **10**, eaam8812 (2017).
- Krasteva, P. V. & Sondermann, H. Versatile modes of cellular regulation via cyclic dinucleotides. *Nat. Chem. Biol.* **13**, 350–359 (2017).
- Burdette, D. L. et al. STING is a direct innate immune sensor of cyclic di-GMP. *Nature* **478**, 515–518 (2011).
- Sun, L., Wu, J., Du, F., Chen, X. & Chen, Z. J. Cyclic GMP–AMP synthase is a cytosolic DNA sensor that activates the type I interferon pathway. *Science* **339**, 786–791 (2013).
- Kranzusch, P. J. et al. Structure-guided reprogramming of human cGAS dinucleotide linkage specificity. *Cell* **158**, 1011–1021 (2014).
- Davies, B. W., Bogard, R. W., Young, T. S. & Mekalanos, J. J. Coordinated regulation of accessory genetic elements produces cyclic di-nucleotides for *V. cholerae* virulence. *Cell* **149**, 358–370 (2012).
- Hu, D. et al. Origins of the current seventh cholera pandemic. *Proc. Natl Acad. Sci. USA* **113**, E7730–E7739 (2016).
- Dziejman, M. et al. Comparative genomic analysis of *Vibrio cholerae*: genes that correlate with cholera endemic and pandemic disease. *Proc. Natl Acad. Sci. USA* **99**, 1556–1561 (2002).
- Severin, G. B. et al. Direct activation of a phospholipase by cyclic GMP–AMP in *El Tor Vibrio cholerae*. *Proc. Natl Acad. Sci. USA* **115**, E6048–E6055 (2018).
- Hornung, V., Hartmann, R., Ablasser, A. & Hopfner, K.-P. OAS proteins and cGAS: unifying concepts in sensing and responding to cytosolic nucleic acids. *Nat. Rev. Immunol.* **14**, 521–528 (2014).
- Jean, S. S., Lee, W. S., Chen, F. L., Ou, T. Y. & Hsueh, P. R. *Elizabethkingia meningoseptica*: an important emerging pathogen causing healthcare-associated infections. *J. Hosp. Infect.* **86**, 244–249 (2014).
- Jenal, U., Reinders, A. & Lori, C. Cyclic di-GMP: second messenger extraordinaire. *Nat. Rev. Microbiol.* **15**, 271–284 (2017).
- Corrigan, R. M. & Gründling, A. Cyclic di-AMP: another second messenger enters the fray. *Nat. Rev. Microbiol.* **11**, 513–524 (2013).
- Woodward, J. J., Iavarone, A. T. & Portnoy, D. A. c-di-AMP secreted by intracellular *Listeria monocytogenes* activates a host type I interferon response. *Science* **328**, 1703–1705 (2010).
- Wang, C. et al. Synthesis of all possible canonical (3′–5′-linked) cyclic dinucleotides and evaluation of riboswitch interactions and immune-stimulatory effects. *J. Am. Chem. Soc.* **139**, 16154–16160 (2017).
- McFarland, A. P. et al. Sensing of bacterial cyclic dinucleotides by the oxidoreductase RECON promotes NF- κ B activation and shapes a proinflammatory antibacterial state. *Immunity* **46**, 433–445 (2017).
- Burroughs, A. M., Zhang, D., Schäffer, D. E., Iyer, L. M. & Aravind, L. Comparative genomic analyses reveal a vast, novel network of nucleotide-centric systems in biological conflicts, immunity and signaling. *Nucleic Acids Res.* **43**, 10633–10654 (2015).
- Kazlauskienė, M., Kostiuk, G., Venclovas, Č., Tamulaitis, G. & Siksnys, V. A cyclic oligonucleotide signaling pathway in type III CRISPR–Cas systems. *Science* **357**, 605–609 (2017).
- Niewoehner, O. et al. Type III CRISPR–Cas systems produce cyclic oligoadenylate second messengers. *Nature* **548**, 543–548 (2017).
- Hallberg, Z. F. et al. Hybrid promiscuous (Hypr) GGDEF enzymes produce cyclic AMP–GMP (3′, 3′-cGAMP). *Proc. Natl Acad. Sci. USA* **113**, 1790–1795 (2016).
- Nelson, J. W. et al. Control of bacterial exoelectrogenesis by c-AMP–GMP. *Proc. Natl Acad. Sci. USA* **112**, 5389–5394 (2015).

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Competing interests Harvard Medical School and the Dana-Farber Cancer Institute have patents pending for CD-NTase technologies on which A.T.W., J.B.E., J.J.M. and P.J.K. are listed as inventors.

Additional information

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METHODS

Bacterial strains and growth conditions. *E. coli* was cultivated at 37 °C, shaking, in LB medium (1% tryptone, 0.5% yeast extract and 0.5% NaCl w/v) and stored in LB plus 30% glycerol at –80 °C unless otherwise indicated. When appropriate, carbenicillin (100 µg ml^{–1}), ampicillin (100 µg ml^{–1}) and chloramphenicol (20–34 µg ml^{–1}) were used. BL21 *E. coli* (strain CodonPlus (DE3)-RIL transformed with pRARE2, Agilent) was used for all protein expression and DH10β *E. coli* (strain Top10, Invitrogen) was used for cloning and plasmid propagation. For repression of protein expression from pET vectors, BL21 *E. coli* was cultivated in MDG medium (0.5% glucose, 25 mM Na₂HPO₄, 25 mM KH₂PO₄, 50 mM NH₄Cl, 5 mM Na₂SO₄, 2 mM MgSO₄, 0.25% aspartic acid and trace metals) with ampicillin and chloramphenicol. For optimum protein expression from pET vectors, BL21 *E. coli* was cultivated in M9ZB medium (0.5% glycerol, 1% Cas-amino acids, 47.8 mM Na₂HPO₄, 22 mM KH₂PO₄, 18.7 mM NH₄Cl, 85.6 mM NaCl, 2 mM MgSO₄ and trace metals) with ampicillin and chloramphenicol²⁷.

Cloning and plasmid construction. Cloning and plasmid construction were performed as previously described²⁸. In brief, for vectors constructed in this study, genes were either amplified from genomic DNA or synthesized as gBLOCKs (Integrated DNA Technologies) with ≥18 base pairs of homology flanking the insert sequence and ligated into a restriction endonuclease linearized vector by Gibson assembly. Reactions were transformed into electrocompetent DH10β cells and selected with appropriate antibiotic plates. Sanger sequencing confirmed each vector was free of mutations within the multiple cloning site. N-terminal 6×His-MBP tag and 6×His-SUMO2 tag fusions were constructed using custom pET16MBP²⁹ or pETSUMO2³⁰ vectors, respectively. cGAS standards used *Homo sapiens* cGAS, DisA standards used *Bacillus thuringiensis* disA and WspR standards used *Pseudomonas aeruginosa* wspR with a D70E constitutively activating mutation³¹. CD-NTases and their effector coding sequences were codon-optimized for bacterial expression (Integrated DNA Technologies) with the exception of genes derived from *E. coli* strain ECOR31³² (ATCC 35350) and *V. cholerae* strain C6706. Synthases were overexpressed in mammalian cells from pcDNA4 plasmids as previously described³³. For expression of MBP N-terminally tagged *dncV* and *cdnE* in mammalian cells, MBP and the fused CD-NTase were codon-optimized for expression in human cells. For coexpression with their putative effector genes, N-terminal MBP-tagged CD-NTases were cloned into pBAD33³⁴ modified with a ribosomal binding site and oriT for conjugation. For cloned CD-NTase details see Supplementary Table 2 and for cloned CD-NTase effector details see Supplementary Table 3.

Recombinant protein purification. Proteins were purified as previously described³⁰. In brief, chemically competent BL21 *E. coli* was transformed with a protein-expression plasmid, recovered on MDG plates overnight, cultivated as a 30 ml starter culture in MDG liquid medium overnight at 37 °C with 230 r.p.m. shaking and used to seed an M9ZB culture at approximately 1:1,000. Then, 25 ml or two 1-l M9ZB cultures were cultivated for approximately 5 h at 37 °C with 230 r.p.m. shaking until the optical density (OD) at 600nm was 2–3.5 at which time cells were chilled on ice for 20 min, IPTG was added at 0.5 mM and cultures were shifted to 16 °C with shaking 230 r.p.m. overnight. Harvested *E. coli* cultures were washed in 1× PBS and stored as a flash-frozen pellet at –80 °C or cultures were immediately disrupted by sonication in lysis buffer (20 mM HEPES-KOH pH 7.5, 400 mM NaCl, 30 mM imidazole, 10% glycerol and 1 mM DTT). Lysates were clarified by centrifugation, filtered through glass wool and proteins were purified by affinity chromatography using Ni-NTA (Qiagen) resin and a gravity column. Resin was washed (lysis buffer supplemented to 1 M NaCl), eluted (lysis buffer supplemented to 300 mM imidazole) and the eluate was dialysed overnight at 4 °C (20 mM HEPES-KOH pH 7.5, 300 mM NaCl and 1 mM DTT). For SUMO2 fusion proteins, dialysis was supplemented with approximately 250 µg of human SENP2 protease (D364–L589, M497A)³⁵. Small-scale preparations of proteins were flash-frozen at this stage and stored at –80 °C in storage buffer (10% glycerol, 20 mM HEPES-KOH pH 7.5, 250 mM KCl and 1 mM TCEP). Where appropriate, proteins were filter-concentrated using centrifugation and a 10-kDa or 30-kDa cut-off column (Millipore Sigma).

For large-scale protein preps (cGAS, DncV, DisA, CdnE, RmCdnE, EmCdnE, STING, RECON, LpCdnE02, EcCdnD02 and CyaA), size-exclusion chromatography followed by concentration was performed in storage buffer without glycerol. Initial CD-NTase proteins were purified and screened as N-terminal MBP fusions to increase expression and stability. Proteins were either freshly thawed from –80 °C stocks and immediately used or maintained at –20 °C in a storage buffer with 50% total glycerol. Glycerol stocks of CD-NTases stored at –20 °C retain >90% activity for more than 6 months and were used for biochemical assays. **Biochemistry and nucleotide synthesis assays.** Recombinant CD-NTase reactions combined: 4 µl of 5× reaction buffer (250 mM CAPSO pH 9.4, 175 mM KCl, 25 mM Mg(OAc)₂ and 5 mM DTT), 2 µl of 10× NTPs, 1 µl [α -³²P]NTPs (approximately 1 µCi), 1 µl of candidate enzyme in storage buffer (approximately 20 µM) and a remaining volume of nuclease-free water for a total reaction volume of 20 µl.

The final reactions (50 mM CAPSO pH 9.4, 50 mM KCl, 5 mM Mg(OAc)₂, 1 mM DTT, ≤5% glycerol, 25–250 µM individual NTPs, trace amounts of [α -³²P]NTP and 1 µM enzyme) were started with addition of enzyme. Where indicated, the pH was altered by replacing the CAPSO buffer with the appropriate buffer from a StockOptions pH Buffer Kit (Hampton Research). When appropriate, Mg²⁺ was replaced with an equimolar concentration of Mn²⁺ (MnCl₂). cGAS reactions were carried out with Tris at pH 7.5 and supplemented with 1 µM ISD45 dsDNA³⁶. Reactions were carried out with 25 µM of each indicated NTP for Figs. 1b, 2c, 4b, c and Extended Data Figs. 1d, 2a–c, 4c, 6c–f, 7a–c, 9d. Reactions in all other figures were carried out with 250 µM NTPs. The NTP/[α -³²P]NTPs in Fig. 1b are cGAS (ATP, GTP/[α -³²P]GTP), DncV (ATP, GTP/[α -³²P]ATP), DisA (ATP/[α -³²P]ATP), WspR (GTP/[α -³²P]GTP) and CdnE (NTP/[α -³²P]NTP). Nuclease P1-treated reactions in Extended Data Figs. 1d, 7c are cGAS (ATP, GTP/[α -³²P]ATP), DncV (ATP, GTP/[α -³²P]GTP), CdnE (ATP, UTP/[α -³²P]ATP) and LpCdnE02 (CTP, UTP/[α -³²P]UTP). Where indicated, non-hydrolysable nucleotides (Ap(c)pp, Gp(c)pp, Cp(c)pp or Up(n)pp (Jena Bioscience)) were used at 25 µM.

Reactions were incubated for 2 h at 37 °C before analysis unless otherwise stated. Reactions were stopped by addition of 5 U of alkaline phosphatase (New England Biolabs), which removed triphosphates on remaining NTPs and converted the remaining nucleotide α -³²P to ³²P_i enabling the visualization of cyclized species. After a ≥20-min incubation, 0.5 µl (PEI–cellulose) or 1 µl (silica) of the reaction was spotted 1.5 cm from the bottom of the TLC plate, spaced 0.8 cm apart. When migration of P_i appeared inconsistent (for example, due to variability in pH) samples were diluted tenfold in 100 mM sodium acetate pH 5.2. The 20-cm × 20-cm F-coated PEI–cellulose TLCs (Millipore) were developed in 1.5 M KH₂PO₄ (pH 3.8) until the buffer front reached approximately 1 cm from the top; 20-cm × 10-cm F-coated silica HP-TLC plates (Millipore) were developed in 11:7:2 1-propanol:NH₄OH:H₂O in a chemical fume hood for 1 h. Plates were dried and exposed to a phosphor screen before detection by a Typhoon Trio Variable Mode Imager system (GE Healthcare).

Nucleotide synthesis and purification. Cyclic dinucleotides and oligonucleotides were produced in large scale using previously described methods³⁷ with the following changes. Small-scale nucleotide synthesis assays were scaled up to 10–40-ml reactions with final conditions of 50 mM CAPSO pH 9.4, 12.5–50 mM K/NaCl, 5–20 mM Mg(OAc)₂, 1 mM DTT, ≤5% glycerol, 250 µM individual NTPs and 1 µM enzyme. A 20-µl aliquot of the larger reaction was removed and [α -³²P]NTPs were added to monitor reaction progress. Reactions were incubated for 24 h at which time 5 U ml^{–1} of alkaline phosphatase (New England Biolabs) was added and the reaction was further incubated for 2–24 h. Reactions were heat-inactivated at 65 °C for 30 min, diluted to a final salt concentration of 12.5 mM and purified by anion exchange chromatography (either 1 ml Q-sepharose column or Mono Q 4.6/100 PE, GE Healthcare). The column was washed with water and 1 ml fractions were collected during a gradient elution with 2 M ammonium acetate. Fractions containing the appropriate product were identified by absorbance at 260 nm (A₂₆₀) and silica TLC, visualizing the nucleotide products by ultraviolet-light shadowing, imaging using a handheld camera and comparing migration to paired, radiolabelled reactions detected by phosphorimaging. Selected fractions were concentrated by evaporation and resuspended in 30 µl of nuclease-free water for mass spectrometry. For NMR, nucleotides were further purified using size chromatography (Superdex 30 Increase 10/300 GL, GE Healthcare). The column was equilibrated with H₂O running buffer and 1 ml fractions were collected, identified by A₂₆₀, pooled and evaporated. Concentrations of purified nucleotides were estimated from A₂₆₀ using the estimated extinction coefficients based on RNA oligonucleotides: cUMP–AMP, ϵ = 22,800 l mole^{–1} cm^{–1}; cAAG, ϵ = 37,000 l mole^{–1} cm^{–1}.

Mass spectrometry. Liquid chromatography coupled with electrospray ionization-mass spectrometry (LC/ESI-MS) analysis was performed using an Agilent 6530 QTOF mass spectrometer coupled to a 1290 infinity binary LC system operating the electrospray source in positive ionization mode. All samples were chromatographed on an Agilent ZORBAX Bonus-RP C18 column (4.6 mm × 150 mm; 3.5 µm particle size) at 50 °C column temperature. The solvent system consisted of 10 mM ammonium acetate (A) and methanol (B). The HPLC gradient with a flow rate of 1 ml min^{–1} starts at 5% B, holds for 2 min and then increases over 12 min to 100% B. Identification of CDNs and cAAG was performed by targeted mass analysis for exact masses and formulae for all possible CDNs and cAAG using Profinder software (version B.06.00 build 6.0.606.0, Agilent).

NMR. All NMR experiments were conducted on a Varian 400-MR spectrometer (9.4 T, 400 MHz). Samples were prepared by resuspending evaporated nucleotide samples in 500 µl D₂O supplemented with 5 mM TMSP (3-(trimethylsilyl)propionic-2,2,3,3-*d*₄) at 27 °C. Data were processed and figures were generated using Vnmrj software (version 2.2C). ¹H and ³¹P chemical shifts are reported in parts per million (p.p.m.). *J* coupling constants are reported in units of frequency (hertz) with multiplicities listed as s (singlet), d (doublet) and m (multiplet). These data appear in the figure legends of each NMR spectrum.

Phospholipase assay. Patatin-like lipases were assayed as previously described³⁸. In brief, CapV and CapE were produced recombinantly and catalytic activity was measured using the EnzChek Phospholipase A1 Assay Kit (Invitrogen) according to the manufacturer's instructions. Phospholipases (250 nM) were incubated with 2.5, 0.25 or 0.025 μ M CDN. c-di-AMP (Invivogen), 3',3'-cGAMP (Invivogen) and c-di-GMP (Biolog) were purchased as chemical standards, cUMP-AMP was purified as described above. Assays were monitored fluorometrically (excitation, 460 nm; emission, 515 nm) for 60 min at approximately 90-s intervals at room temperature using a Biotek Synergy plate reader. Slope of each reaction in the linear range was used to calculate activity (linear regression/straight line analysis, GraphPad Prism 7.0c). A PLA1 standard curve from 20 to 0.02 U was used to interpolate phospholipase activity. Emission was monitored at a gain of 100 and/or 50 in order to extend the linear range of the assay.

Crystallization and structure determination. CdnE homologues were crystallized in apo form or in complex with nucleotide substrates at 18°C using hanging-drop vapour diffusion. Purified *RmCdnE* and *EmCdnE* were diluted on ice to 7–10 mg ml⁻¹ and used immediately to set trays. Alternatively, co-complex crystals were grown by first incubating *RmCdnE* and *EmCdnE* in the presence of a total concentration of combined nucleotides of approximately 10 mM and 10.5 mM MgCl₂ on ice for 30 min. RECON–cAAG co-complex crystals were grown by pre-incubating approximately 10 mg ml⁻¹ RECON (K68A, K70A) with 1 mM of purified *EcCdnD02* product. Following incubation, 2 μ l hanging drops were set at a ratio of 1:1 or 1.2:0.8 (protein:reservoir) over 350 μ l of reservoir in Easy-Xtal 15-Well trays (Qiagen). Optimized crystallization conditions were as follows: apo *RmCdnE*, 100 mM Tris-HCl pH 7.5, 10–20% ethanol; *RmCdnE*–Apcpp–Upnp, 0.24 M sodium malonate, 24% PEG-3350; apo *EmCdnE*, 21 mM sodium citrate pH 7.0, 100 mM HEPES-KOH pH 7.5, 16% PEG-5000 MME; *EmCdnE*–GTP–Apcpp, 100 mM tri-sodium citrate pH 6.4, 10% PEG-3350; *EmCdnE*–pppApA, 100 mM tri-sodium citrate pH 7.0, 8% PEG-3350; RECON–cAAG, 0.1 M NaOAc, 1.0 M LiCl, 30% PEG-6000. Crystals grew in 3–30 days, and all crystals were harvested using reservoir solution supplemented with 10–25% ethylene glycol using a nylon loop, except apo *RmCdnE* crystals, which were harvested using NVH oil. X-ray diffraction data were collected at the Advanced Light Source (beamlines 8.2.1 and 5.0.1) and the Advanced Photon Source (beamlines 24-ID-C and 24-ID-E).

Data were processed with XDS and AIMLESS³⁹ using the SSRL autoxos script (A. Gonzalez, Stanford SSRL). Experimental phase information for *RmCdnE* was determined using data collected from crystals grown with selenomethionine-substituted protein as previously described¹⁰. Four sites were identified with HySS in PHENIX⁴⁰, and an initial map was calculated using SOLVE/RESOLVE⁴¹. Model building was completed in Coot⁴² before refinement in PHENIX. Following model completion, the apo *RmCdnE* structure was used for molecular replacement to determine the nucleotide-bound structures. *RmCdnE* models were not sufficient to phase *EmCdnE* data, but a minimal core *RmCdnE* active-site model was able to successfully determine the substructure and assist experimental phasing with data collected from a native crystal using sulfur single-wavelength anomalous dispersion at a minimal accessible wavelength (approximately 7,235 eV). The 16 heavy sites were identified in HySS that correspond to 12 sulfur and 4 phosphate sites in the *EmCdnE*–pppApA structure, and *EmCdnE* model building was completed as for *RmCdnE*. RECON–cAAG data were phased by molecular replacement using the previously determined RECON–c-di-AMP structure (PDB 5UXF²¹), and model building was manually completed in Coot. X-ray data for refinement were extended according to I/σ resolution cut-off of approximately 1.5 and CC* correlation and R_{pin} parameters. Final structures were refined to stereochemistry statistics for Ramachandran plot (favoured/allowed), rotamer outliers and MolProbity score as follows: *RmCdnE* apo, 98.6%/1.4%, 0.4% and 0.98; *RmCdnE*–Apcpp–Upnp, 98.9%/1.1%, 0.4% and 1.25; *EmCdnE* apo, 97.8%/2.2%, 0.8% and 1.08; *EmCdnE*–GTP–Apcpp, 97.8%/2.2%, 0.8%, and 1.05; *EmCdnE*–pppApA, 98.1%/1.9%, 0.8%, and 1.29; RECON–cAAG, 99.7%/0.3%, 0.7%, and 1.23. See Supplementary Table 1 and 'Data availability' for deposited PDB codes.

Structural comparisons. Structure-based comparisons used to define the conserved CD-NTase architecture in Fig. 2d were calculated using the Dali server and enzymes were clustered according to z -score⁴³ of the core NTase domain. Structures used include: DncV (4TYO¹⁰), cGAS (6CTA³⁰), OAS1 (4RWO⁴⁴), Poly(A) polymerase γ (PAP, 4LT6⁴⁵), CCA-adding enzyme (4X4T⁴⁶), Pol- β (4KLQ⁴⁷) and Pol- μ (4YD1⁴⁸).

In the text and in figures, side-chains are numbered according to the *RmCdnE* sequence. The analogous residue to Asn166 from *RmCdnE* in *E. coli* CdnE is Asn174, in *EmCdnE* is Ser169, in DncV is Ser259 and in human cGAS is Ser378. RECON crystal structures with cAAG, c-di-AMP (5UXF²¹) and NAD (3LN3) were superimposed using PyMOL (version 1.7.4.4) and all structure figures were prepared using PyMOL.

Gel shift assays. In vitro binding assays were performed as previously described³³. In brief, recombinant *Mus musculus* STING or RECON, at 4, 20 or 100 μ M was incubated with radiolabelled nucleotides (≤ 1 μ M final concentration) in gel shift

buffer for final conditions of 50 mM Tris pH 7.5, 60 mM KCl, 5 mM Mg(OAc)₂ and 1 mM DTT. Experiments were prepared by combining 1 μ l of α^{32} P-labelled nucleotides, 2 μ l 5 $\times$ gel shift buffer, 5 μ l of nuclease-free water and started by addition of 2 μ l of recombinant protein in storage buffer. α^{32} P-labelled nucleotides were produced with 25 μ M of each NTP and approximately 1 μ Ci of each [α^{32} P]NTP in the following conditions: cGAS (ATP, GTP/[α^{32} P]GTP), DncV (ATP, GTP/[α^{32} P]ATP), DisA (ATP/[α^{32} P]ATP), WspR (GTP/[α^{32} P]GTP), CdnE (ATP, UTP/[α^{32} P]UTP) and *EcCdnD02* (ATP, GTP/[α^{32} P]GTP).

After 30 min of equilibration, bound and free nucleotides were separated by 6% native PAGE in 0.5 $\times$ TBE buffer, gels were dried and exposed to a phosphorscreen before detection by a Typhoon Trio Variable Mode Imager system (GE Healthcare). Gel shifts were quantified with ImageQuant 5.2 and the percentage of bound nucleotides was calculated as a proportion of total bound and free nucleotides for each lane, after subtraction of background signal.

Cellular assays for interferon- β induction. In-cell assays were performed as previously described³³. In brief, HEK293T cells were transfected using Lipofectamine 2000 in 96-well format with: a control plasmid constitutively expressing *Renilla* luciferase (2 ng pRL-TK), a reporter plasmid expressing interferon- β inducible firefly luciferase (20 ng), a plasmid expressing *M. musculus* STING (5 ng) and a fivefold dilution series of pcDNA4-based plasmids expressing a nucleotidyltransferase (1.2, 6, 30 and 150 ng). 2',3'-cGAMP was produced with human cGAS, 3',3'-cGAMP was produced with *V. cholerae* DncV, cyclic di-AMP (cAA) was produced with *Bacillus subtilis* DisA, cyclic di-GMP (cGG) was produced with *P. aeruginosa* WspR. Luciferase production was quantified after 24 h and firefly luciferase was normalized to *Renilla*, which was then normalized to empty nucleotidyltransferase vector used at 150 ng.

Cell lines. HEK293T cells were used to measure a reporter plasmid expressing interferon- β inducible firefly luciferase. Cell lines were originally provided by the ATCC, no methods were used for authentication, and cell lines were not tested for mycoplasma.

Western blot analysis. CD-NTase in-cell expression levels were verified by western blot of lysed cells. Confluent HEK293T cells were seeded 24 h before transfection at a dilution of 1:4 in a six-well dish. Cells were transfected with 2 μ g of plasmid using Lipofectamine 2000. At 24 h post-transfection, cells were collected by washing cells from the dish using Hanks' buffered saline solution, pelleted at low speed and flash-frozen. Pelleted cells were lysed by resuspending the pellet in 400 μ l 1 $\times$ LDS buffer (ThermoFisher Scientific) and 5% β -mercaptoethanol, boiling for 5 min and vigorously vortexing. Samples were separated by SDS–PAGE, transferred to a nitrocellulose membrane and probed with primary antibodies 1:5,000 rabbit anti-MBP (Millipore AB3596, RRID:AB_91531) and 1:10,000 mouse anti-tubulin (Millipore MABT205, RRID:AB_11204167), followed by secondary antibodies at 1:10,000 IRDye 680RD goat anti-rabbit IgG (LI-COR Biosciences 925-68071, RRID:AB_2721181) and IRDye 800CW goat anti-mouse IgG (LI-COR Biosciences 925-32210, RRID:AB_2687825). Stained membranes were imaged using a LI-COR Odyssey CLx imager.

RECON enzyme assay. Activity assays were performed as previously described²¹. In brief, a twofold dilution series of nucleotides from 50 to 0.05 μ M was incubated in 1 $\times$ PBS with 200 μ M NADPH (co-substrate) and 25 μ M 9,10-phenanthrenequinone (substrate). The reactions were started with the addition of RECON to a final concentration of 0.5 μ M and absorbance at 340 nm was monitored at 20-s intervals for 20 min. The slope of each reaction in the linear range (20–250 s) was used to calculate activity (linear regression/straight line analysis, Prism 7.0c). Values were normalized to reactions with no CDN/nucleotide signalling molecule added, which defined 100% activity. Nucleotides were produced or purchased as described above and cAAG was purified as described above.

Bioinformatics and tree construction. To bioinformatically map CD-NTase-like enzymes in bacteria, we extended a previous analysis²² that combined iterative BLAST analysis, secondary structure predictions and hidden Markov models to collect DncV-like proteins and their genomic context. We identified homologues of each of these 1,300 identified proteins by BLAST analysis of the NCBI non-redundant protein database, then combined these datasets to identify more than 5,600 CD-NTase-like genes. The dataset was then manually curated (Geneious Software) according to shared CD-NTase structural and active-site features. Bacterial genomes and sequences were aligned using MAFFT FFT-NS-2 algorithm version 7.388⁴⁹, a BLOSUM62 scoring matrix, an open gap penalty of 2 and an offset value of 0.123. Proteins with large truncations or lacking the essential DNA polymerase β -like nucleotidyltransferase residues [that is, GS...(D/E)X(D/E)...(D/E)] were removed. The tree was generated from the MAFFT alignment using a Jukes–Cantor genetic distance model, neighbour-joining method, no outgroup and resampled by bootstrap for 100 replicates sorted by topologies. The unrooted tree is used to represent global CD-NTase diversity and probably does not accurately reflect the specific evolutionary relationship between the major CD-NTase A–H clades. The aligned sequences along with pairwise identity comparisons were extracted and used to define clades and clusters. A cluster was defined as more

than 10 CD-NTases that share over 24.5% identity to the sequence preceding each in the alignment. For clarity, 14 poorly aligned CD-NTases were excluded from the tree and are indicated in Supplementary Table 2. The full dataset organized by order from the alignment and containing pairwise comparison of protein identity to each preceding gene is available as Supplementary Table 2 and as Source Data for Fig. 4a. For details on identifying CD-NTases within an organism of interest or organisms encoding a CD-NTase of interest, see Supplementary Discussion.

Each sequence was identified from the nonredundant database of protein sequences and, at times, represents identical proteins translated from genes found in multiple bacteria. For this reason, additional metadata were extracted for each sequence from the NCBI identical protein groups (IPG) database. The number of 'protein accessions' in IPG was used as a surrogate quantification of the number of isolated bacterial genomes that contain each CD-NTase. At the time of access (3 February 2018), 5,686 nonredundant CD-NTases sequences were identified representing a total of 16,717 genomes. At that time, 130,135 bacterial genomes had been deposited in the NCBI Genome database, leading to the crude approximation of 12.8% of genomes containing CD-NTase genes. As some of these genomes may have more than one CD-NTase and the IPG database can overestimate the number of genomes encoding a given protein, we have estimated that more than 10% of bacterial genomes sequenced encode CD-NTases. Taxonomic analysis was performed using metadata associated with each nonredundant CD-NTase record in NCBI and when multiple bacteria were represented by one identical sequence the highest common taxonomical group was used. IPG and taxonomic data can also be found in Supplementary Table 2.

Type CD-NTase enzymes were manually selected from clusters based on the relevance of the organism from which they were isolated (that is, human or plant pathogen/commensal organism), their predicted aptness for in vitro expression (thermophilic organisms or isolates from *E. coli*), the similarity of their operon to the DncV/CdnE operons, and the number of identical protein sequences represented by each unique sequence. CD-NTase001 was selected as and additional control and is encoded by *dncV_{E. coli}* in ECOR31⁵⁰.

CD-NTase screen. Each type CD-NTase gene was codon-optimized for *E. coli*, synthesized (IDT), cloned as an N-terminal 6×His-MBP-fusion and the protein was purified from a 25-ml culture. *E. coli* growth, protein induction and bacterial disruption were performed as described above. Lysates were clarified by centrifugation and Ni-NTA affinity purification was performed as described above with gravity columns replaced by spin columns, centrifuged at 100g. Buffer exchange of eluted proteins was performed by concentrating the eluate using a 0.5-ml 10-kDa cut-off spin column (Ambion) followed by dilution with storage buffer and reconcentration 3× (final imidazole concentration, approximately 0.3 mM). Proteins were analysed for nucleotide synthesis fresh and flash-frozen for storage at −80°C. For biochemical screens, ATP/CTP/GTP/UTP were used at 25 μM each and incubated overnight with the reaction conditions indicated using methods described above. Then, 1 μl of screened protein (approximately 1 μg) was added to the reaction and the same volume assessed by SDS–PAGE followed by Coomassie staining, shown in Extended Data Fig. 6g.

Figure 4d was manually constructed based on known TLC migration patterns that guided CDN identification in each sample. The quantity of ions detected by mass spectrometry relative to other CD-NTases was used to determine whether products were a major or minor constituent. On PEI–cellulose, cyclic dipurines migrate similarly, cyclic purine–pyrimidine hybrids migrate similarly and cyclic dipyrimidines migrate similarly; on silica c-di-AMP migrates uniquely, cyclic UMP–AMP and cGAMP migrate similarly, c-di-GMP and c-di-UMP migrate similarly, and cGMP–UMP and cCMP–UMP migrate similarly; these cannot be distinguished.

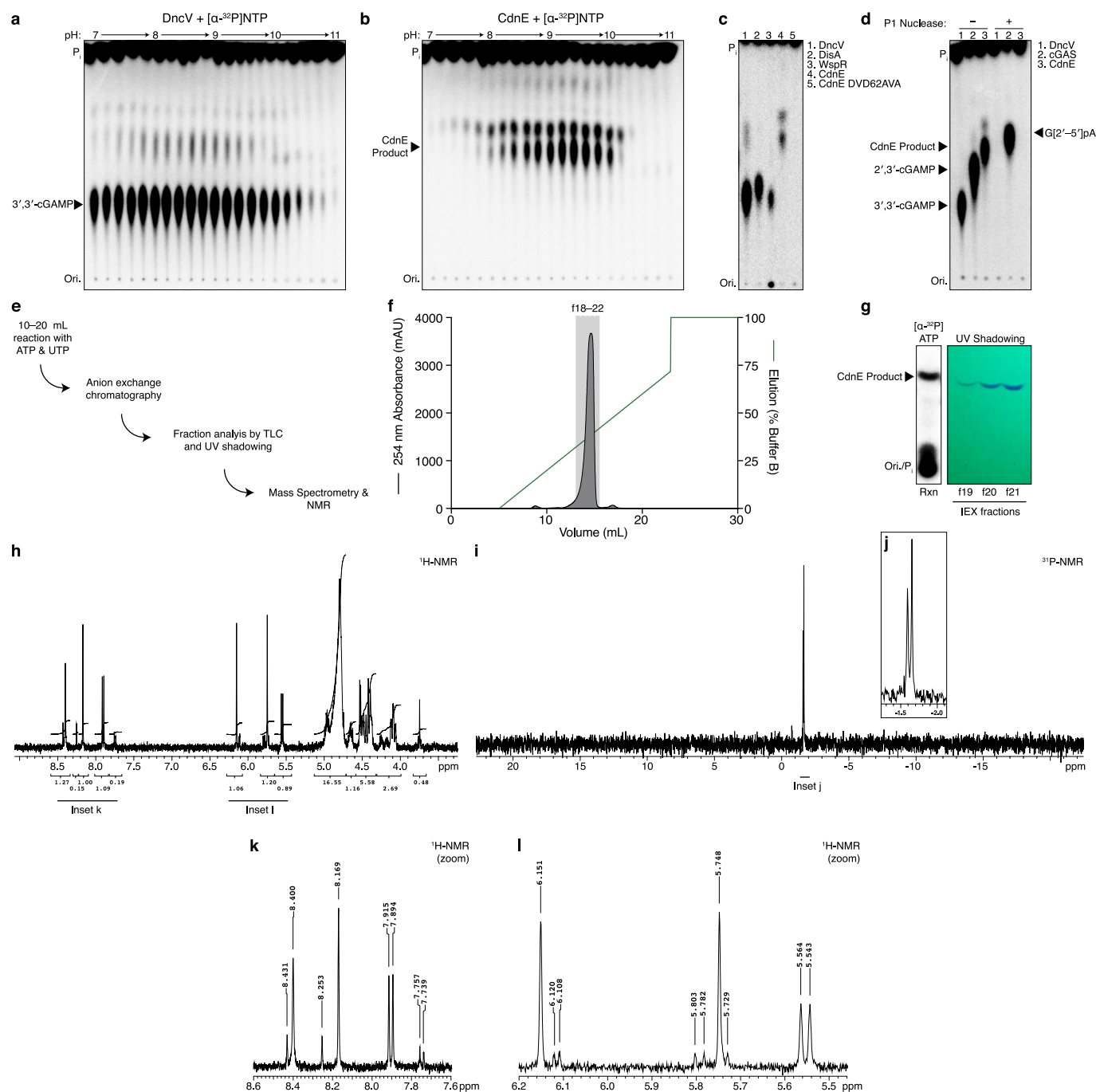
Coexpression of CD-NTases and effectors in *E. coli*. CD-NTases chosen for in-depth analysis were cloned into an arabinose-inducible, chloramphenicol-resistant pBAD33 plasmid. Putative CD-NTase effector genes were selected based on proximity, if they were classified as involved in biological conflict²² and based on analogous operon architecture to known effector phospholipases. Effector genes were codon-optimized for *E. coli* and cloned into pETSUMO2³⁰, a carbenicillin-resistant vector that is IPTG inducible in BL21-DE3 *E. coli* (ThermoFisher Scientific). Three pairs of vectors were assessed for each CD-NTase–effector pair: (1) cognate CD-NTase and effector, (2) CD-NTase and GFP, (3) mCherry and effector. Fluorescent proteins were used as negative controls. Vectors were co-transformed into electrocompetent BL21-DE3 *E. coli*, selected with both relevant antibiotics and maintained under non-inducing conditions (0.2% glucose). Overnight bacterial cultures were serially diluted into LB and 5 μl was spot-plated on selective medium containing 5 μM IPTG and 0.2% arabinose. Colony formation was quantified and images were taken at approximately 24 h.

Reporting summary. Further information on research design is available in the Nature Research Reporting Summary linked to this paper.

Data availability

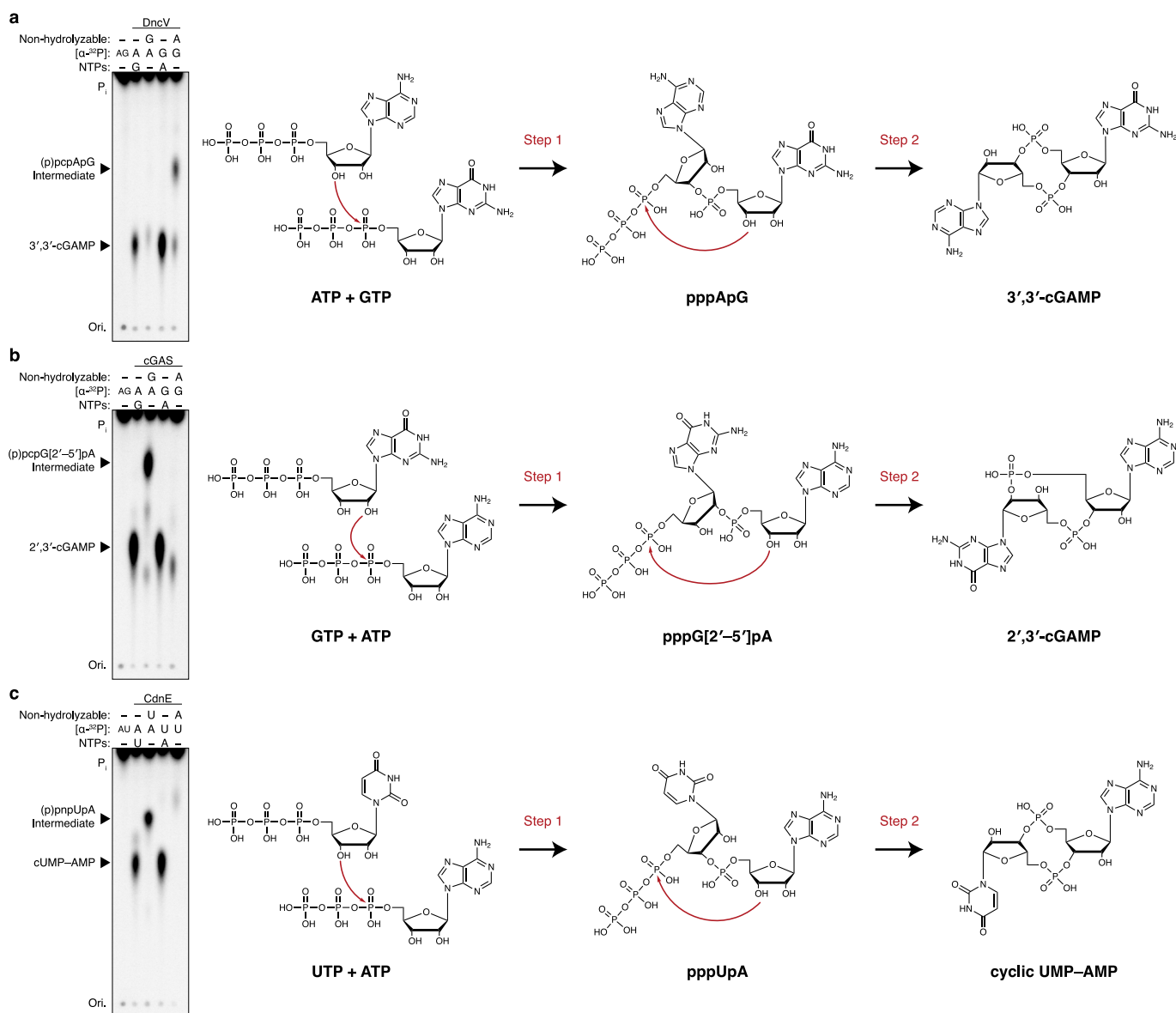
All data supporting the findings of this study are available within the Article and associated Supplementary Information. X-ray crystallographic coordinates and structure factor files are available from the PDB: RmCdnE apo (6E0K); RmCdnE–Apcpp–Upnpp (6E0L); EmCdnE apo (6E0M); EmCdnE–GTP–Apcpp (6E0N); EmCdnE–pppA[3′–5′]pA (6E0O); RECON–cAAG (6M7K). CD-NTase sequences and information on CD-NTase-encoding bacteria are available in Supplementary Table 2 and CD-NTase effector genes sequences are available in Supplementary Table 3. CD-NTase alignments for tree construction are provided as Source Data for Fig. 4a and Source Data are also available for Extended Data Fig. 8b in the online version of the paper. Source gel images are available in Supplementary Fig. 1.

- Studier, F. W. Protein production by auto-induction in high density shaking cultures. *Protein Expr. Purif.* **41**, 207–234 (2005).
- Whiteley, A. T. et al. c-di-AMP modulates *Listeria monocytogenes* central metabolism to regulate growth, antibiotic resistance and osmoregulation. *Mol. Microbiol.* **104**, 212–233 (2017).
- Kranzusch, P. J. & Whelan, S. P. J. Arenavirus Z protein controls viral RNA synthesis by locking a polymerase–promoter complex. *Proc. Natl Acad. Sci. USA* **108**, 19743–19748 (2011).
- Zhou, W. et al. Structure of the human cGAS–DNA complex reveals enhanced control of immune surveillance. *Cell* **174**, 300–311 (2018).
- Kulasakara, H. et al. Analysis of *Pseudomonas aeruginosa* diguanylate cyclases and phosphodiesterases reveals a role for bis-(3′–5′)-cyclic-GMP in virulence. *Proc. Natl Acad. Sci. USA* **103**, 2839–2844 (2006).
- Schubert, S., Dufke, S., Sorsa, J. & Heesemann, J. A novel integrative and conjugative element (ICE) of *Escherichia coli*: the putative progenitor of the Yersinia high-pathogenicity island. *Mol. Microbiol.* **51**, 837–848 (2004).
- Kranzusch, P. J. et al. Ancient origin of cGAS–STING reveals mechanism of universal 2′,3′ cGAMP signaling. *Mol. Cell* **59**, 891–903 (2015).
- Guzman, L. M., Belin, D., Carson, M. J. & Beckwith, J. Tight regulation, modulation, and high-level expression by vectors containing the arabinose PBAD promoter. *J. Bacteriol.* **177**, 4121–4130 (1995).
- Reverter, D. & Lima, C. D. Structural basis for SENP2 protease interactions with SUMO precursors and conjugated substrates. *Nat. Struct. Mol. Biol.* **13**, 1060–1068 (2006).
- Stetson, D. B. & Medzhitov, R. Recognition of cytosolic DNA activates an IRF3-dependent innate immune response. *Immunity* **24**, 93–103 (2006).
- Sureka, K. et al. The cyclic dinucleotide c-di-AMP is an allosteric regulator of metabolic enzyme function. *Cell* **158**, 1389–1401 (2014).
- Gaspar, A. H. & Machner, M. P. VipD is a Rab5-activated phospholipase A1 that protects *Legionella pneumophila* from endosomal fusion. *Proc. Natl Acad. Sci. USA* **111**, 4560–4565 (2014).
- Kabsch, W. XDS. *Acta Crystallogr. D* **66**, 125–132 (2010).
- Adams, P. D. et al. PHENIX: a comprehensive Python-based system for macromolecular structure solution. *Acta Crystallogr. D* **66**, 213–221 (2010).
- Terwilliger, T. C. Reciprocal-space solvent flattening. *Acta Crystallogr. D* **55**, 1863–1871 (1999).
- Emsley, P. & Cowtan, K. Coot: model-building tools for molecular graphics. *Acta Crystallogr. D* **60**, 2126–2132 (2004).
- Holm, L. & Laakso, L. M. Dali server update. *Nucleic Acids Res.* **44**, W351–W355 (2016).
- Löhöfener, J. et al. The activation mechanism of 2′–5′-oligoadenylate synthetase gives new insights into OAS/cGAS triggers of innate immunity. *Structure* **23**, 851–862 (2015).
- Yang, Q., Nausch, L. W. M., Martin, G., Keller, W. & Doublé, S. Crystal structure of human poly(A) polymerase gamma reveals a conserved catalytic core for canonical poly(A) polymerases. *J. Mol. Biol.* **426**, 43–50 (2014).
- Kuhn, C.-D., Wilusz, J. E., Zheng, Y., Beal, P. A. & Joshua-Tor, L. On-enzyme refolding permits small RNA and tRNA surveillance by the CCA-adding enzyme. *Cell* **160**, 644–658 (2015).
- Freudenthal, B. D., Beard, W. A., Shock, D. D. & Wilson, S. H. Observing a DNA polymerase choose right from wrong. *Cell* **154**, 157–168 (2013).
- Moon, A. F., Gosavi, R. A., Kunkel, T. A., Pedersen, L. C. & Bebenek, K. Creative template-dependent synthesis by human polymerase mu. *Proc. Natl Acad. Sci. USA* **112**, E4530–E4536 (2015).
- Katoh, K. & Standley, D. M. MAFFT multiple sequence alignment software version 7: improvements in performance and usability. *Mol. Biol. Evol.* **30**, 772–780 (2013).
- Kato, K., Ishii, R., Hirano, S., Ishitani, R. & Nureki, O. Structural basis for the catalytic mechanism of DncV, bacterial homolog of cyclic GMP–AMP synthase. *Structure* **23**, 843–850 (2015).
- Gao, P. et al. Cyclic [G(2′,5′)pA(3′,5′)p] is the metazoan second messenger produced by DNA-activated cyclic GMP–AMP synthase. *Cell* **153**, 1094–1107 (2013).



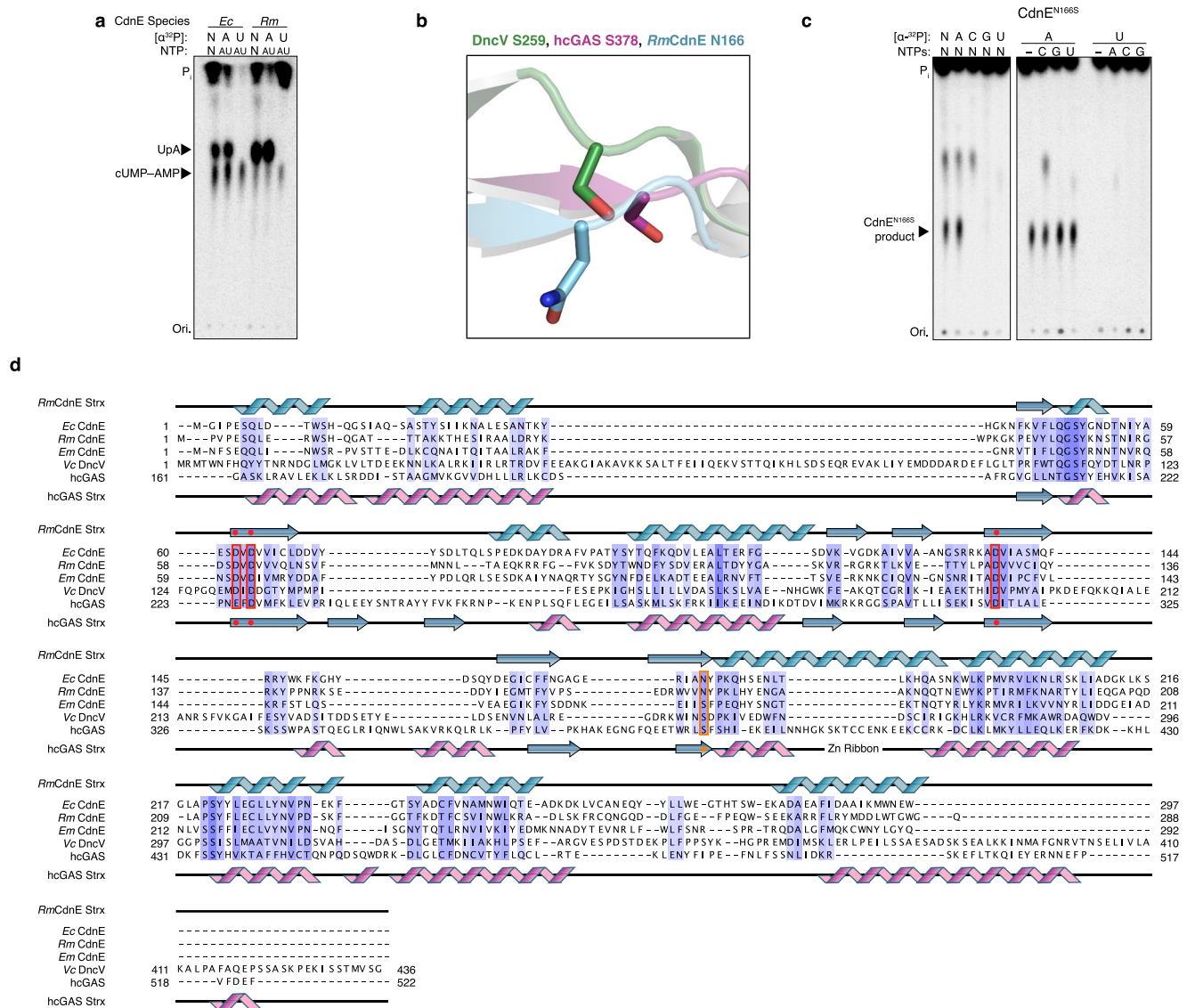
Extended Data Fig. 1 | Detailed characterization of CdnE, a cUMP-AMP synthase. a, b, Titration of reaction buffer pH in steps of 0.2 pH units. Recombinant DncV or CdnE was incubated with α - 32 P-radiolabelled ATP, CTP, GTP and UTP at varying pH and the reactions were treated with alkaline phosphatase and visualized by PEI-cellulose TLC. CdnE activity is optimal at a pH of approximately 9.4, and this reaction condition was used in further experiments. Data are representative of two independent experiments. **c**, PEI-cellulose TLC of products after incubation of the indicated enzyme, wild-type CdnE or active site mutant CdnE with α - 32 P-radiolabelled ATP, CTP, GTP and UTP as in Fig. 1b. Mutations that ablate the CdnE Mg $^{2+}$ -coordinating, active-site residues eliminate all detectable activity. Data are representative of three independent experiments. **d**, Nuclease P1 sensitivity of CDN products. Nuclease P1 specifically cleaves 3'-5', canonical phosphodiester bonds. DncV and CdnE products are completely digested in the presence of P1 and alkaline phosphatase, whereas only one bond of 2',3'-cGAMP is susceptible to digestion, producing the linear G[2'-5']pA product. Data are representative of three

independent experiments. **e**, Workflow of nucleotide production for mass spectrometry and NMR analysis. **f**, Anion-exchange chromatography of a CdnE reaction with ATP and UTP, eluted with a gradient of buffer B (2 M ammonium acetate) by FPLC. Individual fractions were concentrated before pooling for further analysis. **g**, Anion-exchange chromatography (IEX) fractions from **f** were separated by silica TLC, visualized by ultraviolet-light shadowing and compared to a radiolabelled reaction to confirm the appropriate absorbance at 254 nm (A_{254}) peak. Fractions were pooled and concentrated before mass spectrometry and NMR analysis. **h, k, l**, 3',3'-Cyclic uridine monophosphate-adenosine monophosphate proton-NMR spectrum (**h**) and associated magnified spectra (**k, l**) 1 H NMR (400 MHz): δ_{H} 8.40 (s, 1H), 8.17 (s, 1H), 7.90 (d, J = 8.2 Hz, 1H), 6.15 (s, 1H), 5.75 (s, 1H), 5.55 (d, J = 8.2 Hz, 1H), 5.00–4.90 (m, 2H), 4.80 (d, J = 4.5 Hz, 1H) 4.70–4.61 (m, 1H), 4.55–4.38 (m, 4H), 4.17–4.02 (m, 2H). **i, j**, 3',3'-Cyclic uridine monophosphate-adenosine monophosphate phosphate-NMR spectrum (**i**) and associated magnified spectrum (**j**). 31 P{ 1 H} NMR (162 MHz): δ_{P} -1.59 (s, 1P), -1.65 (s, 1P).



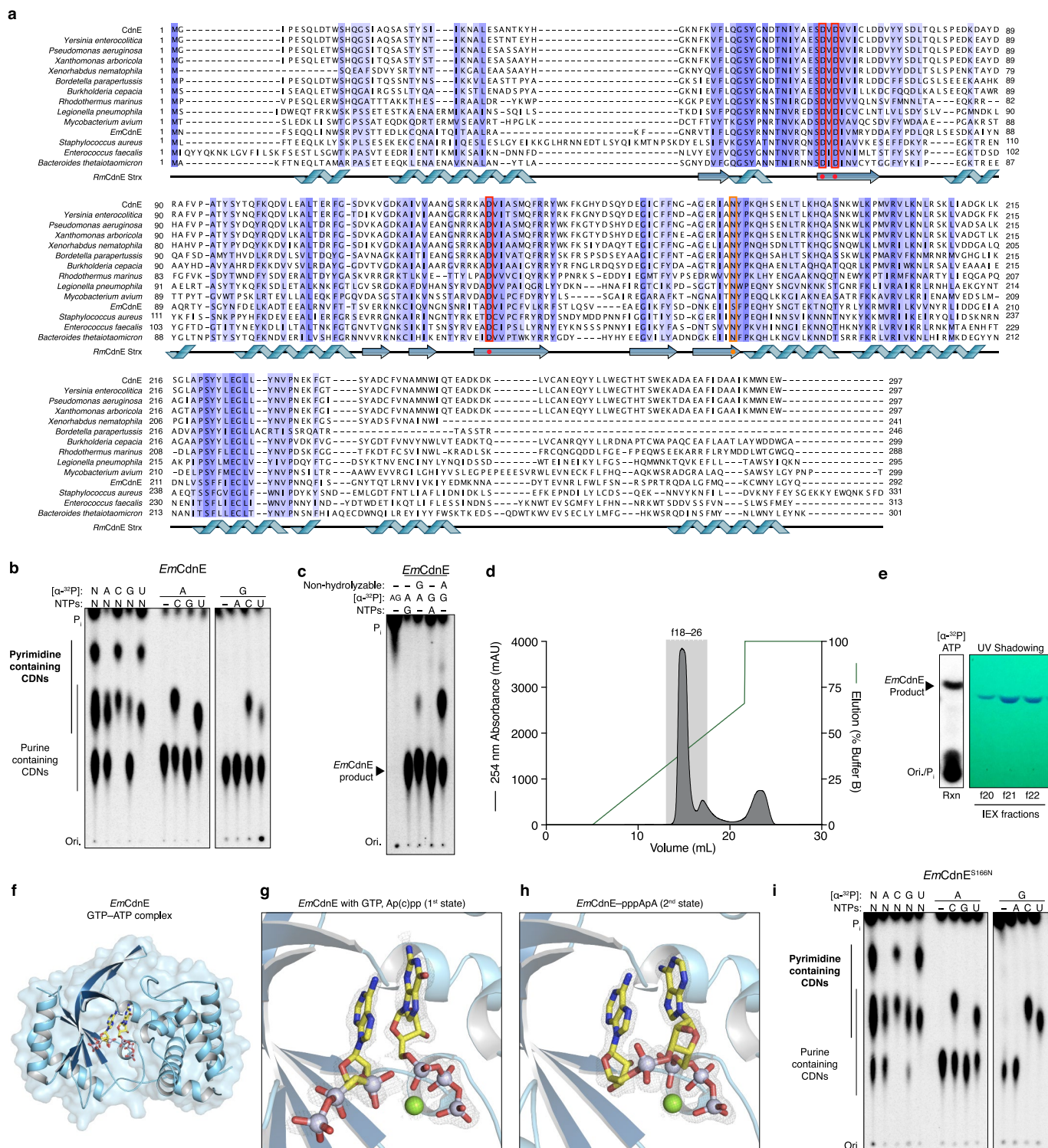
Extended Data Fig. 2 | Order of the DncV, cGAS and CdnE reactions.
a–c, Incubation of CD-NTase enzymes with non-hydrolysable nucleotides traps reaction intermediates and identifies the order of the reaction. Left, PEI–cellulose TLC analysis of reactions as in Fig. 1b, in which individual NTPs have been replaced by their non-hydrolysable nucleotide analogues.

Exposed γ -phosphates are labelled in parentheses to indicate that these are removed by phosphatase treatment before analysis. Data are representative of three independent experiments. Right, previously determined reaction mechanisms (DncV and cGAS)^{10,51} and proposed reaction mechanism for CdnE.



Extended Data Fig. 3 | Detailed analysis of *RmCdnE*. **a**, A thermophilic homologue of CdnE from *R. marinus* (*RmCdnE*) also synthesizes cUMP-AMP. Recombinant proteins were incubated with α^{32} P-radiolabelled NTPs as indicated at either 37°C (*CdnE*) or 70°C (*RmCdnE*) and the reactions were visualized by PEI-cellulose TLC as in Fig. 1b. Data are representative of two independent experiments. **b**, Active site of *RmCdnE* superimposed with structures of cGAS (6CTA³⁰) and DncV (4TYO¹⁰). **c**, The analogous position to Asn166 was mutated in CdnE to a serine and that protein,

CdnE^{N166S}, was characterized further. Reactions, which were analysed as in Fig. 1b, demonstrate that CdnE^{N166S} loses pyrimidine specificity. Data are representative of two independent experiments. **d**, Structure-based sequence alignment of CD-NTases, annotated with secondary structure features of *RmCdnE* and human cGAS (6CTA³⁰). Mg²⁺-coordinating active-site residues are highlighted in red; blue arrows, cyan/pink annotations and purple highlights represent secondary structure; and the analogous residues to *RmCdnE* Asn166 are highlighted in orange.

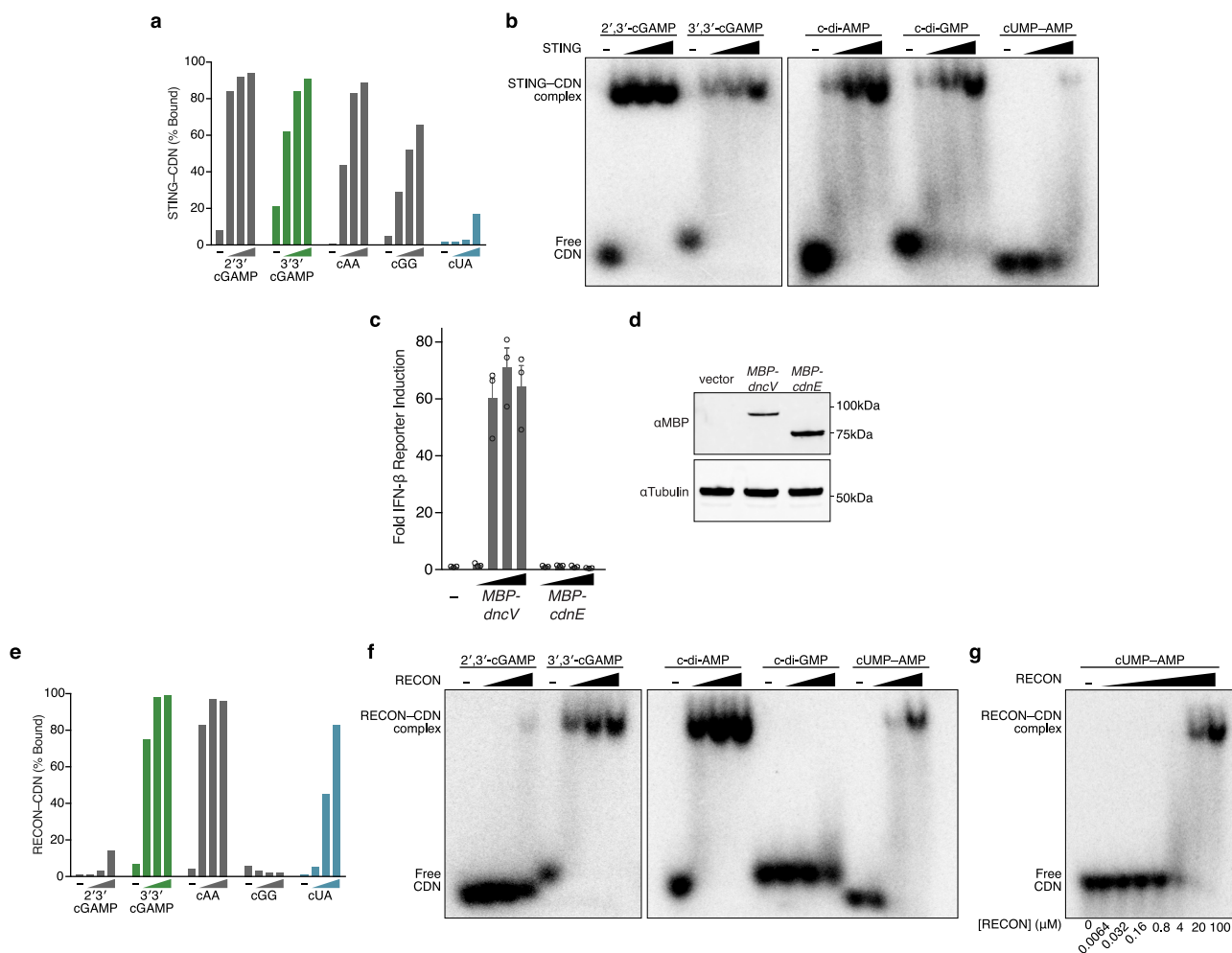


Extended Data Fig. 4 | See next page for caption.

Extended Data Fig. 4 | Detailed structural analysis of *EmCdnE*.

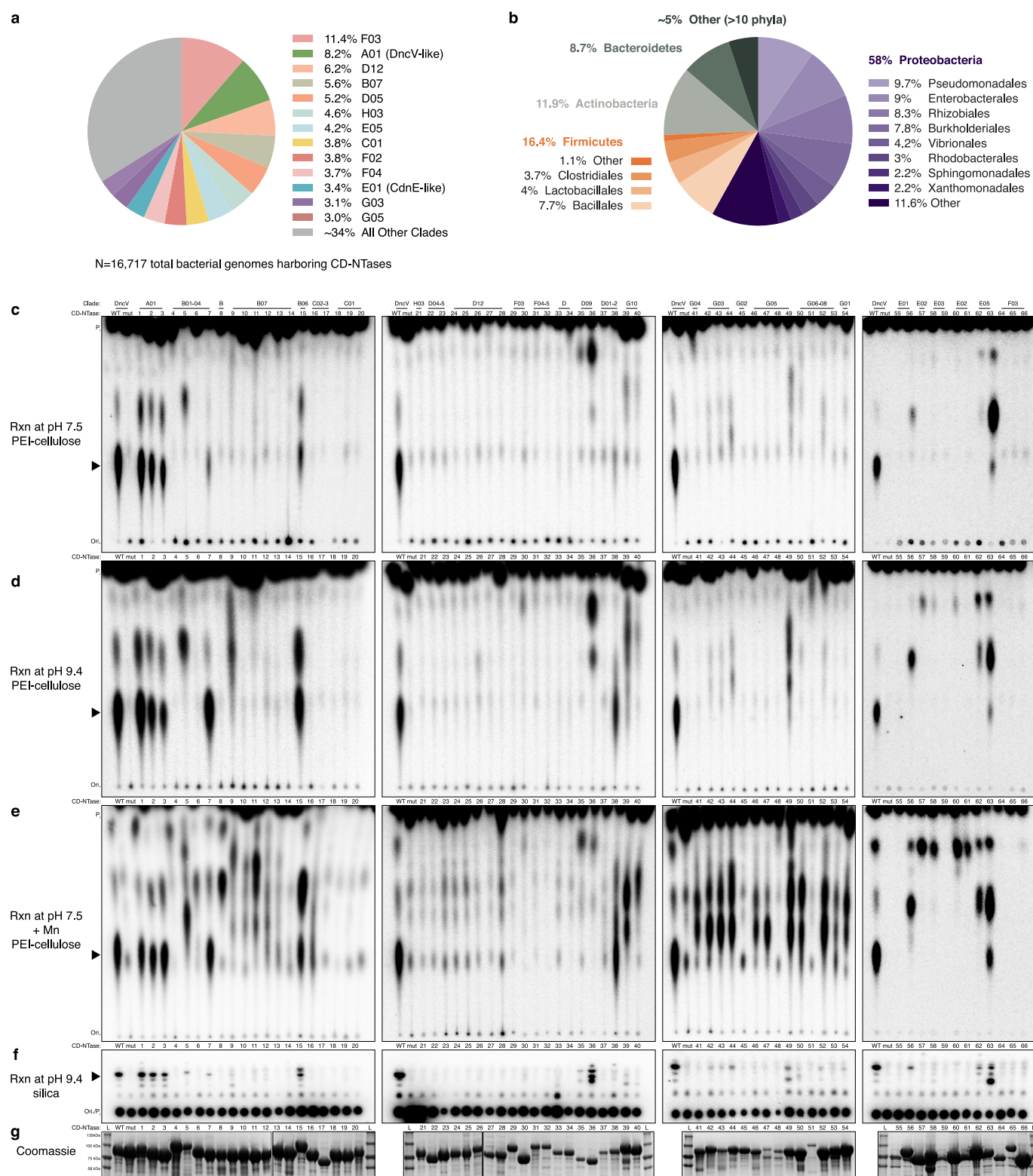
a, Sequence alignment of CdnE homologues in Fig. 2c, annotated with *RmCdnE* secondary structure features. Mg^{2+} -coordinating active-site residues are highlighted in red and the analogous residues to *RmCdnE* Asn166 are highlighted in orange. *Yersinia enterocolitica* (WP_050915017); *P. aeruginosa* (WP_096075289); *Xanthomonas arboricola* (WP_104644370); *Xenorhabdus nematophila* (WP_010848498); *Bordetella parapertussis* (WP_015040391); *Burkholderia cepacia* complex (WP_006482377); *R. marinus* (*RmCdnE*, WP_014072508); *L. pneumophila* (WP_042646516); *Mycobacterium avium* (WP_062886322); *E. meningoseptica* (*EmCdnE*, WP_016200549); *Staphylococcus aureus* (WP_031901603); *Enterococcus faecalis* (WP_050492554); *Bacteroides thetaiotaomicron* (WP_062695386). **b**, Biochemical deconvolution of *EmCdnE*, which has a natural serine substitution at the Asn166 analogous site. Recombinant protein was incubated with NTPs as indicated and reactions were visualized as in Fig. 1b. Data are representative of three independent experiments. **c**, Reactions of *EmCdnE* incubated with $\alpha^{32}P$ -radiolabelled NTPs and non-hydrolysable nucleotide analogues as indicated and visualized as in Fig. 1b. Data are representative of three independent experiments. **d**, Anion exchange chromatography of an *EmCdnE* reaction with ATP and GTP, eluted with a gradient of buffer B (2 M ammonium acetate) by FPLC. Individual fractions were concentrated

before pooling for further analysis. **e**, Anion exchange chromatography fractions from **d** were separated by silica TLC, visualized by ultraviolet-light shadowing and compared to a radiolabelled reaction to confirm the appropriate peak. Fractions were pooled and concentrated before mass spectrometry analysis. Mass spectrometry confirmed synthesis of products with masses corresponding to c-di-AMP, cGAMP and c-di-GMP. **f**, Crystal structure of *EmCdnE* in complex with GTP and non-hydrolysable ATP capturing the 'first state' structure before NTP hydrolysis. Mg^{2+} ions are omitted for clarity. **g**, Magnified cut-away of the active site of the complex shown in **f**, confirming the position of a serine at the analogous site to *RmCdnE* Asn166. Nucleotide and metal $2F_o - F_c$ electron densities are contoured at 1σ . **h**, Magnified cut-away of the active site of the *EmCdnE*-pppApA structure, capturing the 'second state' after the first reaction has occurred to form a linear intermediate, but before CDN formation. Nucleotide and metal $2F_o - F_c$ electron densities are contoured at 1σ . **g, h**, Mg^{2+} ions are shown in green. **i**, Biochemical deconvolution of mutant *EmCdnE* reverted to the ancestral asparagine at the Asn166 analogous site. This mutant loses preference for producing cyclic dipurine molecules and instead produces more pyrimidine-containing CDN products. Reactions were visualized as in Fig. 1b. Data are representative of two independent experiments.



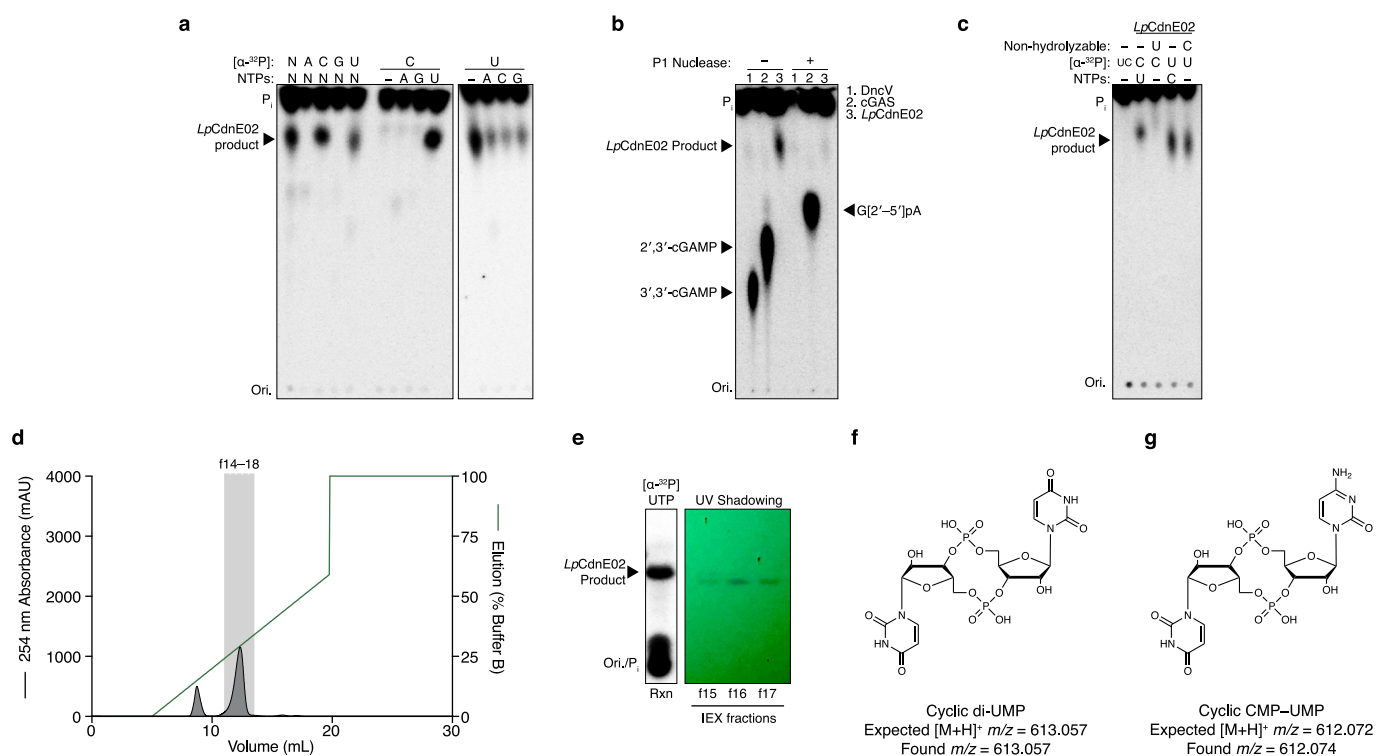
Extended Data Fig. 5 | cUMP-AMP recognition helps to define innate immune receptor specificity. **a, e**, Quantification of nucleotide interactions with the host receptors STING (**a**) or RECON (**e**), measured using a radiolabelled nucleotide bound to a concentration gradient of host protein, separated in a native PAGE gel shift (0, 4, 20, 100 μ M protein). Quantification of the gels shown in **b** and **f**. Data are representative of $n = 2$ independent experiments. **b, f**, Native PAGE gel shift analysis of STING (**b**) or RECON (**f**) complex formation with indicated radiolabelled CDNs. Proteins are titrated at 0 (—), 4, 20 and 100 μ M. STING readily binds all cyclic dipurine species, but does not form a high-affinity complex with cUMP-AMP. RECON readily binds all 3',3'-CDN species that contain at least one adenine base, including cUMP-AMP. Data are representative of two independent experiments. **c**, In-cell STING reporter

assay. Induction of an IFN β reporter in HEK293T cells transfected with a concentration gradient of plasmid-overexpressing enzymes as indicated. DncV and CdnE were expressed with N-terminal MBP tags and IFN β reporter induction was compared as fold over empty vector, shown as (–). Data are mean $\pm$ s.e.m. for $n = 3$ technical replicates and are representative of two independent experiments. **d**, Western blot of MBP-tagged DncV and CdnE expressed from plasmids analysed in **c** to validate in vivo expression. Data are representative of two independent experiments. Gel source data are available in Supplementary Fig. 1. **g**, Gel shift analysis as in **f**, with protein titration to measure the relative affinity of the RECON–cUMP–AMP interaction. Protein concentrations listed below. Data are representative of 2 independent experiments.



Extended Data Fig. 6 | A biochemical screen of CD-NTases from diverse bacterial genera. a, Chart of the number of bacterial genomes ($n = 16,717$) that have CD-NTases from clusters in Fig. 4a. See also Supplementary Table 2. **b,** Taxa of genome-sequenced bacteria isolated with unique CD-NTase genes, phyla are indicated in bold; Proteobacteria and Firmicutes are further divided by order and visualized by shades of colour. **c–f,** Type CD-NTases interrogated for product synthesis.

Purified proteins were incubated with $\alpha^{32}\text{P}$ -radiolabelled NTPs under different reaction conditions (indicated pH and divalent cation) and reaction products were visualized by either PEI-cellulose or silica TLC as in Fig. 1b and Fig. 4c, respectively. **g,** CD-NTase expression level and purity. Coomassie-stained SDS-PAGE analysis of purified CD-NTase enzymes used in each reaction. Data are representative of two independent experiments.

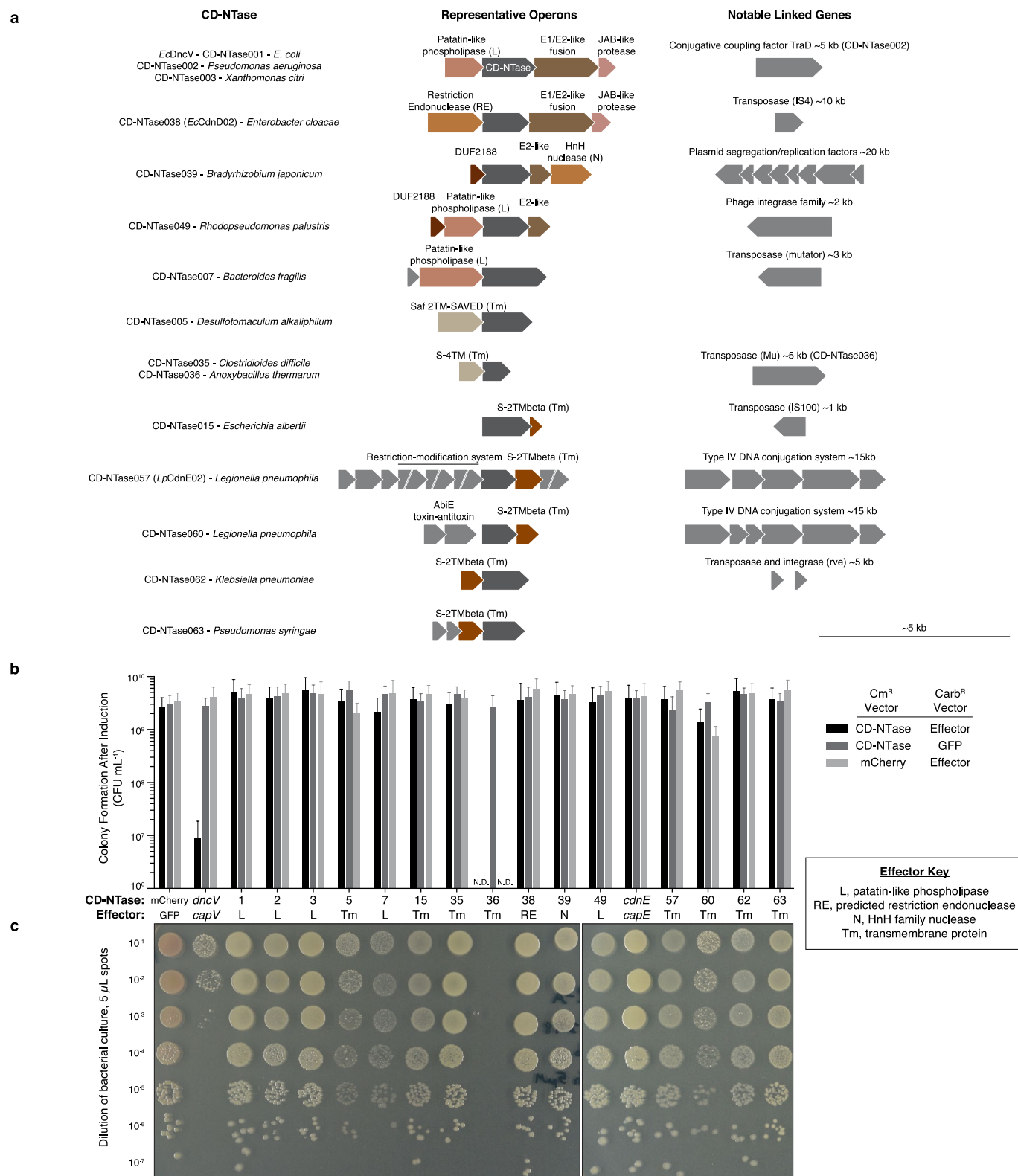


Extended Data Fig. 7 | Detailed biochemical analysis of *LpCdnE02*.

a, Biochemical deconvolution of *LpCdnE02* (CD-NTase057), analysed as in Fig. 1c, demonstrates specific synthesis of cyclic dipyrimidine products. Data are representative of three independent experiments. **b**, Nuclease sensitivity of the *LpCdnE02* product, as described in Extended Data Fig. 1d. Data are representative of three independent experiments. **c**, Incubation of *LpCdnE02* with non-hydrolysable nucleotides, as described in Extended Data Fig. 2. Non-hydrolysable UTP completely blocks the reaction, indicating the first step requires attack of the α P from UTP. However, the product formed when non-hydrolysable CTP is present cannot be distinguished from c-di-UMP in this assay, and it is unclear whether the reaction proceeds through a pppCpU reaction

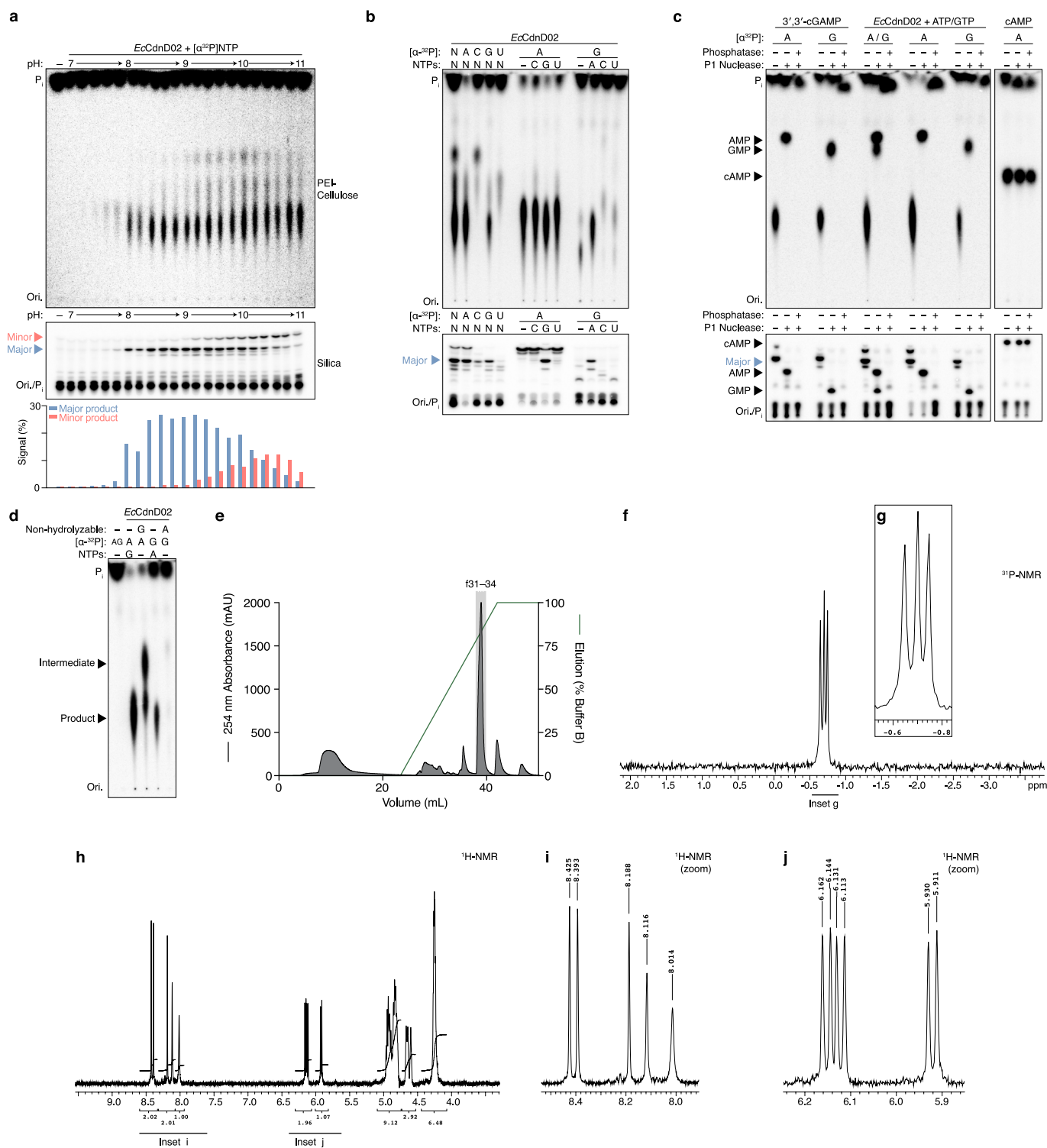
intermediate. Data are representative of three independent experiments.

d, Anion exchange chromatography of a *LpCdnE02* reaction with UTP and CTP, eluted with a gradient of buffer B (2 M ammonium acetate) by FPLC. Individual fractions were concentrated before pooling for further analysis. **e**, Anion exchange chromatography fractions from **d** were separated by silica TLC, visualized by ultraviolet-light shadowing and compared to a radiolabelled reaction to confirm the appropriate peak. Fractions were pooled and concentrated before mass spectrometry analysis. **f**, Mass spectrometry confirmed synthesis of c-di-UMP as the major product of *LpCdnE02*. **g**, Mass spectrometry confirmed synthesis of cCMP-UMP as a minor product of *LpCdnE02*.



Extended Data Fig. 8 | CD-NTases are encoded in conserved, poorly understood operons on mobile genetic elements. **a**, Operon structure for CD-NTases selected for in-depth characterization showing the conserved protein domains in CD-NTase adjacent genes (see Fig. 4a–c). Conserved operons were identified previously and operons are vertically organized by similarity to one another²². Where found, linked genes demonstrating that CD-NTases are encoded on mobile genetic elements are indicated. **b**, CD-NTases and their adjacently encoded ‘effector’ proteins were coexpressed in *E. coli* and bacterial colony formation was quantified by spot-dilution analysis. CD-NTases were inducibly expressed from a chloramphenicol-resistance (Cm^R) vector and effectors were inducibly expressed from a

carbenicillin-resistance (Carb^R) vector. Bacteria that express cognate CD-NTase–effector plasmids or control plasmids were plated on medium containing both inducers and incubated for 24 h at 37°C. Data were not determined (N.D.) for CD-NTase036, because the effector was toxic to *E. coli* under non-inducing conditions. Data are the mean ± s.e.m. of three independent experiments. **c**, Spot-dilution analysis of bacteria expressing the indicated cognate CD-NTase–effector pairs, quantified as in **b**. The CD-NTase036–effector pair was not analysed in this assay. Colony morphology indicates a potential interaction for some combinations; however, it is unclear how specific or meaningful this may be. Data are representative of three independent experiments.

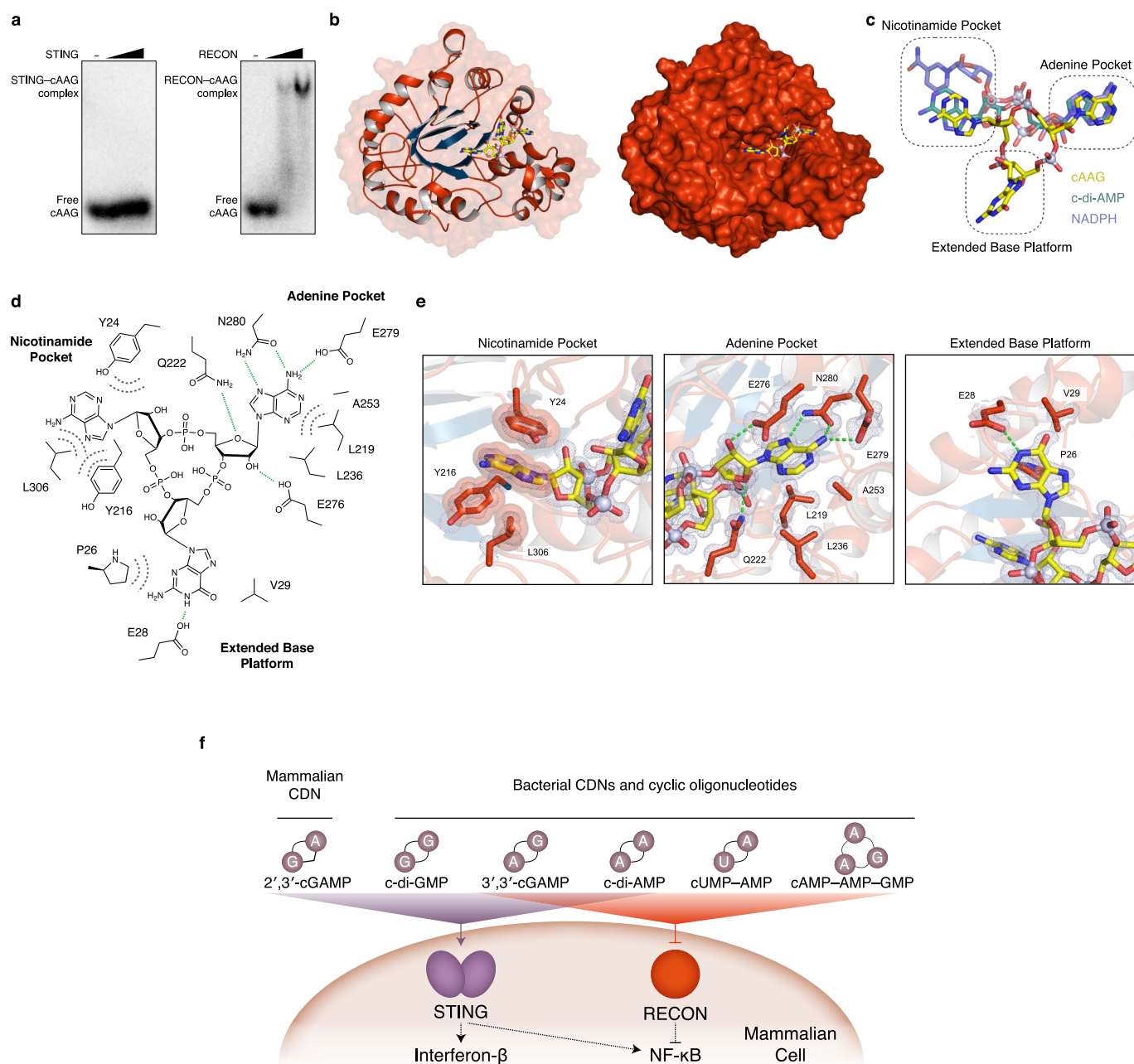


Extended Data Fig. 9 | See next page for caption.

Extended Data Fig. 9 | Detailed biochemical analysis of EcCdnD02.

a, Titration of reaction buffer pH in steps of 0.2 pH units. Recombinant *EcCdnD02* was incubated with $\alpha^{32}\text{P}$ -radiolabelled NTPs at varying pH and the reactions were analysed and visualized by PEI-cellulose or silica TLC as in Fig. 1b and Fig. 4c, respectively. Silica TLC identified two products, denoted the major (blue triangle) and minor (red triangle) product. Quantification of TLC spots is shown at the bottom. Data are representative of two independent experiments. **b**, Biochemical deconvolution of *EcCdnD02*, recombinant protein was incubated with NTPs as indicated and analysed by TLC as in **a**. Data are representative of three independent experiments. **c**, Nuclease digestion of the *EcCdnD02* product. Conventional nuclease digestion includes addition of a phosphatase, in this experiment reactions were first treated with Antarctic phosphatase to remove remaining NTPs and were then heat-inactivated. Next, reactions were untreated, treated with nuclease P1 (specific for 3'-5'-phosphodiester bonds) only or treated with nuclease P1 and phosphatase to remove exposed phosphate groups. 3',3'-cGAMP (DncV) and *EcCdnD02* product are digested into AMP and GMP constituents, which are phosphatase-sensitive. cAMP (CyaA) is insensitive to P1 digestion and cyclic monophosphates are phosphatase-resistant. These data demonstrate that the *EcCdnD02* product does not contain a cyclic monophosphate. Data are representative of three independent

experiments. **d**, Incubation of *EcCdnD02* with non-hydrolysable nucleotides, as described in Extended Data Fig. 2. Non-hydrolysable ATP completely blocks the reaction, indicating the first step requires attack of the αP from ATP. However, when non-hydrolysable GTP is present the possible intermediates (pp(c)pGpA, pp(c)pGpApA, or pppApA) cannot be distinguished in this assay and it is unclear how the reaction proceeds. Silica TLC is not suited for analysing non-hydrolysable nucleotides, because they do not migrate beyond the origin. Data are representative of three independent experiments. **e**, Anion exchange chromatography of a *EcCdnD02* reaction with ATP and GTP, eluted with a gradient of buffer B (2 M ammonium acetate) by FPLC. Individual fractions were concentrated before pooling for further analysis. **f**, **g**, 3',3',3'-Tricyclic adenosine monophosphate-adenosine monophosphate-guanosine monophosphate-phosphate-NMR spectrum (**f**) and associated magnified spectrum (**g**). $^{31}\text{P}\{^1\text{H}\}$ NMR (162 MHz): δ_{P} -0.65 (s, 1P), -0.70 (s, 1P), -0.75 (s, 1P). **h-j**, 3',3',3'-Tricyclic adenosine monophosphate-adenosine monophosphate-guanosine monophosphate proton-NMR spectrum (**h**) and associated magnified spectra (**i**, **j**). ^1H NMR (400 MHz): δ_{H} 8.43 (s, 1H), 8.39 (s, 1H), 8.19 (s, 1H), 8.12 (s, 1H), 8.01 (s, 1H), 6.15 (d, $J = 7.0$ Hz, 1H), 6.12 (d, $J = 7.0$ Hz, 1H), 5.92 (d, $J = 7.5$ Hz, 1H), 5.00-4.78 (m, 6H), 4.69-4.58 (m, 3H), 4.3-4.2 (m, 6H).



Extended Data Fig. 10 | Structural analysis of cAAG inhibition of RECON. **a**, cAAG interactions with STING or RECON, radiolabelled nucleotides incubated with a concentration gradient of each protein, separated in a native PAGE gel shift (0, 4, 20, 100 μ M protein). Data are representative of two independent experiments. **b**, Co-crystal structure of the RECON-cAAG complex shown as cartoon (left) and surface (right). **c**, Overlay and orientation of RECON ligands cAAG, c-di-AMP (PDB 5UXF), co-substrate NAD (PDB 3LN3) demonstrate three individual

binding pockets. **d**, Schematic representation of residues from RECON that interact with cAAG. Green dotted lines indicate hydrogen bonds, grey dotted lines indicate hydrophobic interactions. **e**, Magnified cutaways of individual RECON binding pockets as in **d**. **f**, Mammalian innate-immune sensors recognize CD-NTase products with overlapping specificities. 2',3'-cGAMP and c-di-GMP are detected by STING; 3',3'-cGAMP and c-di-AMP are detected by both STING and RECON; and cUMP-AMP and cAMP-AMP-GMP are detected by RECON.

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- ☐ ☒ The exact sample size (n) for each experimental group/condition, given as a discrete number and unit of measurement
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- ☒ ☐ For null hypothesis testing, the test statistic (e.g. F , t , r) with confidence intervals, effect sizes, degrees of freedom and P value noted
Give P values as exact values whenever suitable.
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- ☐ ☒ Clearly defined error bars
State explicitly what error bars represent (e.g. SD, SE, CI)

Our web collection on [statistics for biologists](#) may be useful.

Software and code

Policy information about [availability of computer code](#)

Data collection

Profinder version B.06.00 build 6.0.606.0 (Agilent), VnmrJ version 2.2C NMR software

Data analysis

Phenix 1.13-2998, Coot 0.8.9, PyMOL 1.7.4.4, Geneious v11, Prism 7, VnmrJ, Profinder version B.06.00 build 6.0.606.0 (Agilent)

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Atomic coordinates and structure factors for the reported crystal structures are deposited at the Protein Data Bank (PDB) under accession codes 6E0K, 6E0L, 6E0M, 6E0N, 6E0O, and 6M7K.

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Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	N/A
Data exclusions	N/A
Replication	All experiments were performed with independent replicates as described in the figure legends.
Randomization	N/A
Blinding	N/A

Reporting for specific materials, systems and methods

Materials & experimental systems

n/a	Involved in the study
☐	☒ Unique biological materials
☐	☒ Antibodies
☐	☒ Eukaryotic cell lines
☒	☐ Palaeontology
☒	☐ Animals and other organisms
☒	☐ Human research participants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Unique biological materials

Policy information about [availability of materials](#)

Obtaining unique materials All unique CD-NTase enzyme genes and described plasmids are available from the authors.

Antibodies

Antibodies used	1:5,000 Rabbit anti-MBP (Millipore Cat# AB3596, RRID:AB_91531); 1:10,000 Mouse anti-Tubulin (Millipore Cat# MABT205, RRID:AB_11204167); 1:10,000 IRDye 680RD Goat anti-Rabbit IgG (LI-COR Biosciences Cat# 925-68071, RRID:AB_2721181); 1:10,000 IRDye 800CW Goat anti-Mouse IgG (LI-COR Biosciences Cat# 925-32210, RRID:AB_2687825)
Validation	Appropriate negative controls and target validation as stated by the manufacturers.

Eukaryotic cell lines

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Cell line source(s)	HEK293T were originally obtained from the ATCC.
Authentication	No methods were used for authentication.
Mycoplasma contamination	The cell lines were not tested for mycoplasma.

Commonly misidentified lines
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No commonly misidentified cell lines were used in this study.